Healthwise Lifespan Assessment Using IoT and AI

A Graduate Project Report submitted to Manipal Academy of Higher Education in partial fulfilment of the requirement for the award of the degree of

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In

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Submitted by

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CERTIFICATE

This is to certify that the project titled **Healthwise Lifespan Assessment Using IoT and AI** is a record of the bonafide work done by **HARDIK SEN** (*Reg. No. 210907288*) submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology (BTech) in **ELECTRONICS AND COMMUNICATION ENGINEERING** of Manipal Institute of Technology, Manipal, Karnataka, (A Constituent Unit of Manipal Academy of Higher Education), during the academic year 2024 - 2025.

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ABSTRACT

In today's world, where sedentary lifestyles and stress have become increasingly common, the need for preventive healthcare has never been more urgent. This project, titled Healthwise Lifespan Assessment Using IoT and AI, aims to bridge technology and health by offering a smart solution for predicting life expectancy and delivering personalized wellness guidance. By integrating Internet of Things (IoT) technologies with Artificial Intelligence (AI), the system allows users to monitor their health status and receive data-driven recommendations tailored to their daily habits and physical condition.

At the core of the system is a machine learning model (XGBoost) trained on a diverse dataset comprising lifestyle and health-related parameters such as age, gender, diet, exercise, BMI, medical history, sleep, alcohol consumption, stress levels, smoking habits, and social life. The model predicts estimated life expectancy based on user input. Alongside this, a domain-specific knowledge base was constructed using a variety of health and wellness books and resources. These documents were embedded and indexed using Pinecone to support fast and intelligent information retrieval, enhancing the quality of personalized recommendations through Retrieval-Augmented Generation (RAG) techniques using the Mistral-7B-Instruct-v0.3 language model.

The project's IoT component plays a crucial role in ensuring real-time physiological health monitoring. Sensors such as the MAX30100 (for heart rate and SpO2) and MLX90614 (for body temperature) are connected to an ESP32 microcontroller. This hardware setup streams real-time sensor readings to Firebase using Wi-Fi and HTTP protocols. These live values are automatically fetched and displayed in the web form, eliminating manual input and making the system more seamless, accurate, and user-friendly. The real-time integration allows dynamic health status tracking, which adds another layer of precision to the AI-generated health insights.

The entire pipeline—from data collection and prediction to recommendation and interaction—is hosted on a web platform powered by FastAPI and a responsive HTML/CSS/JavaScript frontend. The platform begins by showing the predicted life expectancy and the AI-generated personalized advice, after which users can continue the conversation with the chatbot. Tools such as LangChain, HuggingFace, Pinecone, and the Mistral model are used to enable natural language communication and memory retention. Overall, this project demonstrates the potential of combining AI, IoT, and modern software frameworks to build intelligent, proactive healthcare systems that promote long-term well-being.

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CHAPTER 1
INTRODUCTION

1.1 Introduction

As the global focus shifts from reactive to preventive healthcare, there is a rising interest in technologies that can monitor, predict, and guide personal well-being. The project titled *Healthwise Lifespan Assessment Using IoT and AI* combines the power of Artificial Intelligence and Internet of Things to offer a forward-looking solution. It aims to not only estimate a user's life expectancy based on lifestyle choices but also provide smart, actionable health suggestions. The motivation behind this work is to bridge the existing gap between isolated fitness apps and intelligent advisory systems by developing a fully integrated platform that understands and evolves with the user.

1.2 Introduction to the Area of Work

The integration of AI and IoT is reshaping healthcare by enabling continuous, context-aware systems that respond to users in real time. Through the use of wearable sensors, machine learning, and conversational AI, it's now feasible to track key health indicators and offer guidance that is both timely and personalized. This has given rise to predictive and preventive health tools that aim to improve lifestyle and extend longevity. Our work sits at the intersection of these innovations—bringing together machine learning, IoT sensor data, cloud-based infrastructure, and language models to create a comprehensive life assessment tool.

1.3 Present Day Scenario

In today's fast-paced world, lifestyle-related illnesses such as obesity, high blood pressure, and cardiovascular conditions are increasingly common. Most people remain unaware of the long-term effects of their daily habits due to the absence of real-time personalized insights. While several health applications exist, they often fall short in offering a unified platform that predicts life expectancy, monitors physiological parameters, and provides dynamic recommendations. This project addresses this shortfall by combining real-time sensor data with AI-based prediction models, making it possible to deliver accurate, actionable, and personalized health advice in one ecosystem.

1.4 Motivation

Existing healthcare solutions are either reactive—focused on treating symptoms after they appear—or generic, offering broad advice without personalization. Meanwhile, AI and IoT technologies have matured enough to deliver meaningful real-time health monitoring and analysis. The availability of low-cost sensors and cloud infrastructure makes it feasible to build smart systems that understand and guide users based on both historical patterns and current biometrics. This project is driven by the need to merge these capabilities into a seamless platform that encourages healthier lifestyles through awareness, prediction, and engagement.

1.4.1 Findings from Literature

Kaur and Singh (2021)[36] discussed the integration of IoT in health systems, emphasizing its potential in real-time data collection but also highlighting major limitations in intelligence and personalization. *Gupta and Pandey (2020)*[40] proposed an IoT-based architecture for monitoring vitals but did not include predictive modeling or adaptive feedback. *Kumar and Sharma (2019)*[41] analyzed ML-based health systems but noted a lack of integration between multiple intelligent layers — such as real-time sensing, prediction, and personalized guidance.

1.4.2 Identification of Gaps to be Addressed

- *Limited Adaptivity*: Prior works use static or rule-based systems that lack personalization or real-time updates.
- *Isolated Functionalities:* Most systems are either data collection tools or diagnostic engines — few integrate prediction and advisory services.
- *Lack of RAG and LLMs:* Retrieval-augmented generation (RAG) and large language models (LLMs) are rarely utilized for contextual, dynamic health recommendations.
- *User Engagement:* Minimal focus on user interactivity, existing platforms lack chatbot-based conversational interfaces for guided health coaching.

1.4.3 How This Work Bridges the Gaps

- *Unified Architecture*: Combines IoT data collection (future scope), ML-based life expectancy prediction (XGBoost), and LLM-driven health advice into a single cohesive pipeline.
- *Dynamic Personalization*: Uses context-aware embeddings and retrieval (via Pinecone) to provide tailored responses powered by the Mistral-7B-Instruct-v0.3 LLM.
- *Conversational Interface*: Introduces a user-centric chat platform where structured recommendations are shared upfront and followed by interactive, personalized Q&A.
- *Real-Time Integration*: Designed to integrate more real-time data (heart rate, SpO2, temperature) over time, ensuring evolving accuracy and engagement.

1.4.4 Significance in Present Context

With the increasing adoption of wearable technology and demand for personalized healthcare, the timing of this work is critical. Users want more than just health data—they want to understand what it means and how to act on it. This project not only empowers users with insights but also encourages healthier choices through intelligent, interactive engagement, making it highly relevant to current healthcare trends.

1.5 Objective of the Work

- Design and implement an AI-based health monitoring system that predicts life expectancy using machine learning models trained on lifestyle and biometric data.
- Enable real-time health tracking by integrating IoT sensors (heart rate, SpO_2, temperature) and connecting them to a web-based platform for continuous data acquisition and analysis.
- Provide personalized, context-aware health recommendations using a large language model (LLM) and semantic retrieval from a curated knowledge base, validated through real-time pilot trials to assess system responsiveness, usability, and effectiveness.

1.6 Target Specifications

- Train an XGBoost model on a rich lifestyle dataset.
- Embed health literature using HuggingFace embeddings and store in Pinecone.
- Use RAG (Retrieval-Augmented Generation) with Mistral-7B-Instruct-v0.3 to provide personalized advice.
- Build a FastAPI backend that receives user input and handles predictions.
- Integrate IoT hardware for real-time vitals monitoring (heart rate, SpO2, temperature).
- Display sensor values directly in a web form synced with Firebase.

1.7 Project Work Schedule

- *Phase 1 (Weeks 1–3):* Finalize scope, perform literature survey, select sensors and tools.
- *Phase 2 (Weeks 4–8):* Collect and clean dataset, test ML models and prepare embeddings.
- *Phase 3 (Weeks 9–12):* Build vector store, integrate with LLM, test health recommendation pipeline.
- *Phase 4 (Weeks 13–16):* Complete full-stack development including chatbot and form UI.
- *Phase 5 (Weeks 17–20):* Develop and test IoT-Firebase integration, ensure accurate sensor reading display.
- *Phase 6 (Weeks 21–24):* Gather user feedback, finalize report, prepare presentations and documentation.

1.8 Organization of the Report

- *Chapter 1*: Introduction, motivation, problem statement, and scope of the work.
- *Chapter 2:* Background theory and literature review covering existing work and technical foundation.
- *Chapter 3:* Methodology covering data pipeline, IoT setup, and AI integration.
- *Chapter 4:* Results, performance analysis, and evaluation of predictions and recommendations.

- *Chapter 5:* Conclusion with project learnings and suggestions for future work.
- *Chapter 6:* Health, Safety, Risk and Environment Aspects, Evaluates potential risks, safety precautions, and environmental considerations.
- *References:* Lists all academic references and resources used in IEEE format.
- *Project Details:* A summary of the project contributors and institutional information.

CHAPTER 2
BACKGROUND THEORY

2.1 Introduction

This chapter lays the theoretical groundwork for the project titled *Healthwise Lifespan Assessment Using IoT and AI*. It explores key technologies like IoT, machine learning, and natural language processing—core pillars of this system. Alongside, it draws insights from recent research and innovations in smart healthcare. The aim is to understand how these technologies work together to build a real-time, predictive, and personalized health advisory system that not only estimates lifespan but also guides users toward healthier living.

2.2 Introduction to the Project Title

The project title *Healthwise Lifespan Assessment Using IoT and AI* reflects a vision that extends beyond just data analysis. It is about building a smart ecosystem where real-time health data collected via IoT sensors feeds into AI models to generate insights. These insights are then presented to the user through an intelligent chatbot that adapts advice based on individual needs. By combining sensor-based vitals (heart rate, SpO_2, temperature), lifestyle inputs, and advanced machine learning, the system aims to give people a clearer picture of their current health and future well-being.

2.3 Literature Review

2.3.1 Present State and Recent Developments

Over the last decade, we've seen an explosion in health tech tools—smartwatches, fitness trackers, mobile health apps—all designed to help people monitor their well-being. These devices can measure everything from pulse to oxygen saturation in real time. Simultaneously, AI techniques like machine learning have become essential for processing and learning from health data, making it possible to predict medical risks or lifestyle impacts with increasing accuracy. At the intersection of these developments lies the potential to create systems that don't just monitor but also guide, recommend, and even converse intelligently with users.

2.3.2 Background Theory

- *Internet of Things (IoT):* IoT refers to a web of interconnected physical devices that collect and exchange data. In healthcare, this means wearable sensors that track vitals and send them to cloud servers or applications. For this project, the ESP32 microcontroller connects to MAX30100 (for heart rate and SpO_2) and MLX90614 (for body temperature), transmitting real-time data to Firebase. This allows seamless data flow from body to browser.
- *Machine Learning:* ML enables systems to learn from data and make predictions. We use the XGBoost model—a high-performance gradient boosting technique—chosen for

its efficiency in handling structured tabular data like age, diet, exercise, stress level, BMI, and more. XGBoost outputs a life expectancy prediction based on user inputs.

- *Natural Language Processing (NLP) and LLMs:* NLP allows machines to understand human language, while LLMs like Mistral-7B-Instruct-v0.3 take it a step further—creating intelligent, context-aware responses. In this project, an LLM delivers personalized health advice through a conversational chatbot, enhancing user engagement.
- *Vector Embeddings and RAG:* Embeddings convert words or documents into numerical vectors. These are stored in Pinecone, a vector database, enabling fast retrieval based on similarity. RAG combines this retrieval step with a generative LLM, resulting in smart and factually relevant responses.

2.3.3 Literature Survey

- *Kumar et al. (2023)* [19]: This study provides a comprehensive review of the current trends, applications, challenges, and security issues in the Healthcare Internet of Things (H-IoT). The authors highlight the integration of H-IoT with technologies like big data, blockchain, machine learning, and edge computing. However, the study points out challenges such as data privacy, scalability, and real-time processing. Our project aims to address these challenges by implementing secure data handling and scalable architectures.

- *A. Kumar & R. Gill (2023)* [20]: The authors discuss the rapid technological changes in healthcare, emphasizing the role of IoT in revolutionizing diagnosis and treatment. They explore the benefits of real-time information sharing and the functional advantages of the Web of Things in healthcare. Nonetheless, the study lacks a detailed exploration of integrating AI for predictive analytics. Our work extends this by incorporating AI-driven predictive models to enhance healthcare delivery.

- *Yassin et al. (2021)* [21]: This survey focuses on the applications of IoT in healthcare and the associated challenges. The authors discuss various IoT applications, including patient monitoring and data collection. They identify challenges like data security, interoperability, and the need for standardized protocols. Our project addresses these issues by implementing secure communication protocols and ensuring interoperability among devices.

- *Gaur (2022)* [22]: The chapter explores the role of IoT in healthcare, discussing its potential to improve patient care and operational efficiency. The author emphasizes the importance of integrating IoT with other technologies for enhanced healthcare services. However, the study does not delve into the challenges of data management and analysis.

Our project focuses on efficient data management strategies and advanced analytics to derive meaningful insights.

- *Kislay et al. (2022)* [23]: This review examines the applications of IoT in healthcare, highlighting benefits such as improved patient monitoring and data collection. The authors discuss challenges like data security, device interoperability, and the need for real-time analytics. Our project aims to overcome these challenges by implementing secure data transmission protocols and real-time data processing capabilities.

- *Ghanavati et al. (2017)* [24]: The authors propose a cloud-assisted IoT-based health status monitoring framework. They discuss the integration of IoT devices with cloud computing to monitor patient health. While the framework offers scalability, it raises concerns about data privacy and latency. Our project addresses these concerns by incorporating edge computing to reduce latency and enhance data privacy.

- *Khan (2024)* [25]: This article discusses the emerging trends and challenges of IoT in smart healthcare systems. The author highlights the integration of IoT with AI and the potential benefits for patient care. However, the study identifies challenges like data security, interoperability, and the need for standardized protocols. Our project focuses on implementing secure and standardized communication protocols to ensure seamless integration.

- *Singh (2024)* [26]: The author explores the opportunities and challenges of IoT in healthcare. The study discusses the potential of IoT to improve patient care and operational efficiency. Nonetheless, it points out challenges such as data privacy, device interoperability, and the need for real-time analytics. Our project aims to address these challenges by implementing secure data handling practices and real-time data processing capabilities.

2.4 Summarized Outcome of Literature Review

The research clearly supports the use of IoT for collecting health vitals and machine learning for predicting outcomes. However, most solutions are siloed—focusing either on data tracking or on predictive analytics, rarely both. Furthermore, very few systems use large language models with vector-based memory to generate personalized advice. Our work stands out by combining health sensing, predictive modeling, and conversational recommendations into a single, interactive platform.

2.5 Theoretical Discussions

- *Health Prediction Modeling:* We use regression-based machine learning (XGBoost) to estimate life expectancy. The model takes structured input features like age, BMI, diet type, exercise, stress, and medical history to produce lifespan predictions.
- *RAG Framework for Recommendations*: The LLM uses Retrieval-Augmented Generation to create responses. First, it searches a knowledge base using vector similarity (via Pinecone), then combines this information with its own generative capabilities to offer personalized advice.
- *IoT in Data Acquisition*: Sensors collect vital signs in real time. This data is sent via WiFi to Firebase, then automatically appears in a web-based health form—creating a seamless experience for users and improving the accuracy of predictions.

2.6 General Analysis

This project combines structured AI modeling with conversational intelligence. While XGBoost offers accurate predictions, the LLM bridges the gap between data and user by generating easy-to-understand recommendations. The use of Firebase and ESP32 ensures that sensor data reaches the cloud reliably and quickly. HuggingFace embeddings and Pinecone provide a powerful retrieval layer, allowing the chatbot to reference accurate medical literature. Together, this hybrid system delivers real-time, data-backed, user-friendly health guidance.

2.7 Conclusions

The theory and research explored in this chapter clearly support the need for and feasibility of an integrated health advisory platform. By bringing together IoT-based sensing, AI-powered predictions, and conversational NLP, this project delivers not just insights—but actionable, personalized, and real-time health recommendations. It opens new doors for proactive, user-centric healthcare systems and lays the foundation for smarter, longer, and healthier living.

CHAPTER 3
METHODOLOGY

3.1 Introduction

This chapter outlines the methodology adopted for developing the *Healthwise Lifespan Assessment Using IoT and AI* system. It details the systematic approach involving dataset creation, machine learning model training, chatbot integration, IoT sensor implementation, and web interface development. The choices of tools, algorithms, and architectural decisions are backed by both technical and clinical relevance.

3.2 Detailed Methodology

The methodology consists of five major stages:

- Dataset Creation & Validation
- Life Expectancy Prediction Model Training (XGBoost)
- Recommendation System (LLM + RAG)
- IoT-Based Health Data Integration
- Web Interface Integration (Frontend +Life Prediction Model + LLM + IoT)
- Final Deployment Flow

Step 1: Dataset Creation & Validation

- Given that no publicly available dataset included the comprehensive set of features required for our analysis — such as age, gender, BMI, sleep duration, physical activity, stress levels, medical history, and social life — we constructed a custom dataset from scratch using MySQL.
- The selection of these features was grounded in scientific research. Each attribute was chosen based on a thorough review of medical literature and global health reports to ensure real-world relevance to life expectancy prediction. The detailed scientific rationale and citations supporting these choices are provided in the References section [1]–[11].
- The step-by-step SQL-based process used to create and modify the dataset — including rule-based logic, transformations, and data structuring — is documented in Annexure A: Dataset Creation and Life Expectancy Adjustment – SQL Workflow.
- To ensure the accuracy and reliability of the dataset, we performed multiple validation checks. This included aligning our data distributions with known medical statistics and ensuring the data followed realistic patterns consistent with clinical observations.
- For full transparency and reproducibility, all data validation, exploratory analysis, and statistical sanity checks can be reviewed in the corresponding GitHub repository, linked in Annexure B: Reproducibility and Model Configuration..

Step 2: Life Expectancy Prediction Model Training (XGBoost)

To predict life expectancy based on lifestyle and health factors, we built a machine learning pipeline using the XGBoost regression algorithm. The entire workflow, as illustrated in Figure 3.1, outlines the journey from raw data to a trained and optimized prediction model.

- *Loading and Preprocessing the Data*

  We began by importing our custom-built dataset from MySQL into Python using Pandas. During this phase, all entries were carefully cleaned — missing values were handled, data formats were standardized, and basic data integrity checks were performed. This initial stage ensured a consistent and usable dataset.

  In Figure 3.1, this phase is represented as the “Raw Data” → “Preprocessing” block.

- *Feature Engineering*

  Once the data was clean, we moved on to feature engineering, where we prepared the I nputs for the model:

  - Categorical features like gender, country, and diet type were converted into machine-readable format using One-Hot Encoding.
  - Numerical features such as BMI, hours of sleep, and exercise hours were standardized using StandardScaler to bring them onto a similar scale.

  This process is visualized in Figure 3.1 where preprocessing branches into two paths:

  → One for categorical features (via One-Hot Encoding)

  → Another for numerical features (via Standard Scaling).

- *Data Splitting*

  The data was then divided into training (80%) and testing (20%) sets. The training set helped the model learn the underlying patterns, while the test set evaluated its performance.

  This step is represented by the “Data Split” block in the figure.

- *Model Optimization with Bayesian Search*

  To fine-tune the model, we implemented Bayesian Hyperparameter Optimization using BayesSearchCV. This method intelligently searched for the best values of hyperparameters like learning rate, tree depth, and number of estimators to maximize the model’s accuracy.

  This optimization step is clearly marked in the workflow under “Bayesian Hyperparameter Optimization”.

- *Pipeline Construction*

  With the preprocessing steps and model structure ready, we assembled them into a pipeline. This modular setup chained preprocessing and training steps together, making the workflow reusable, scalable, and robust to new inputs.

  In Figure 3.1, this is captured under the “Pipeline Construction” block.

- *Model Training and Evaluation*

- We trained the model using the optimized XGBoost regression pipeline and assessed its performance using a comprehensive set of evaluation metrics. The model achieved an R^2 score of 0.89 and a Root Mean Squared Error (RMSE) of approximately 2.1 years, indicating strong predictive capability.

- To further strengthen the evaluation, we included the Mean Absolute Error (MAE), which was found to be 1.68 years, offering a clear measure of the average deviation in predictions. We also conducted 5-Fold Cross-Validation, which yielded an average R^2 score of 0.87 ± 0.03 and an average RMSE of 2.19 ± 0.15 years, demonstrating the model's robustness and stability across different data splits.

- Additionally, we estimated the 95% Confidence Interval for prediction error, which was approximately ± 0.38 years, providing a statistical measure of uncertainty in the model's predictions.

- This entire training and evaluation process is visually summarized in the "Model Training and Evaluation" section of the fig 3.1.

- *Feature Importance Analysis*
  Once the model was trained, we analyzed which features had the most influence on life expectancy predictions. Using XGBoost's built-in feature importance tools, we identified key drivers such as smoking habits, BMI, exercise hours, sleep patterns, and stress levels.
  This corresponds to the "Feature Importance Extraction" block in the figure3.1.
- *Model Export*
  Finally, the trained and validated model pipeline was saved as a .pkl file using Joblib. This allows us to load and use the model later for inference through the web interface.

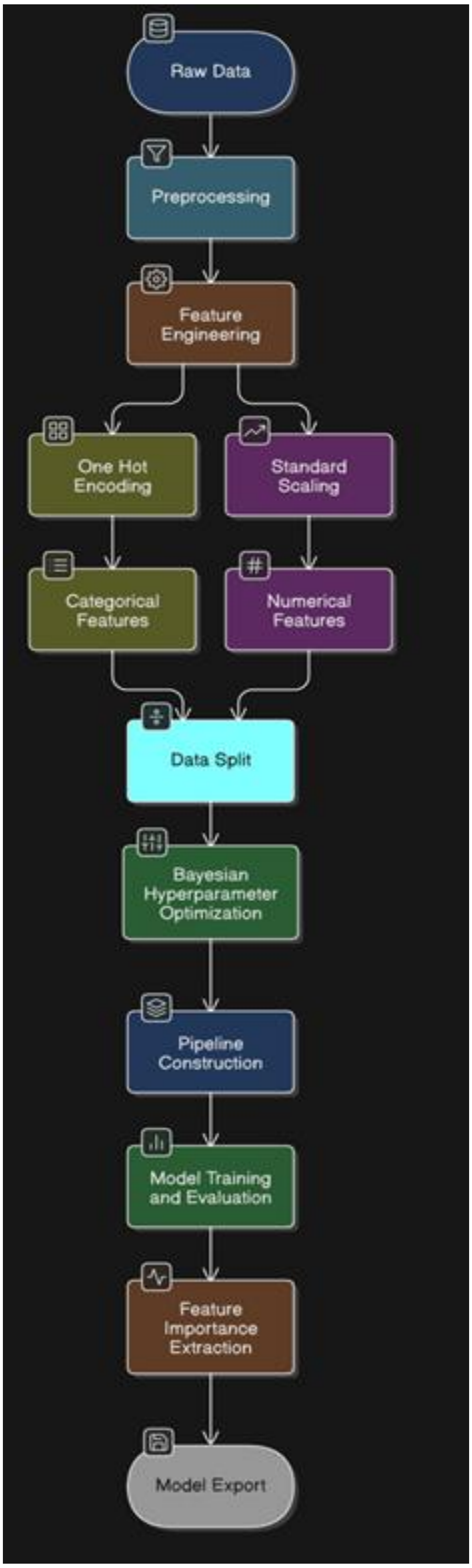

Figure 3.1: Life Expectancy Prediction Model Training Workflow Diagram

Step 3: Recommendation System (LLM + RAG)

While predicting life expectancy is valuable, providing actionable, personalized guidance is equally essential. To achieve this, we developed a Recommendation System that combines the power of Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG), supported by a domain-specific knowledge base. The workflow for this system is illustrated in Figure 3.2.

- *Curating the Knowledge Base*

  The first step was building a high-quality Domain-Specific Knowledge Base (DSKB) focused on health and lifestyle. This knowledge base was meticulously curated using:
  - Books on wellness, nutrition, and preventive care
  - Guidelines from clinical institutions
  - WHO reports and evidence-based health articles

  The aim was to ensure that any advice generated by the AI model is medically relevant and reliable.

  This is the foundational block in Figure 3.2, represented as the "Store Health Sources".
- *Chunking the Text for Efficient Search*

  To make the knowledge easily searchable, all documents were split into smaller text chunks ranging from 100 to 300 tokens. This improves the relevance of search results and ensures that retrieved context fits within the token limits of the LLM.

  In the flow diagram, this step is shown as "Chunk Creation", a crucial pre-processing step that enables precise semantic search.
- *Generating Embeddings with Transformers*

  Each chunk was then converted into a high-dimensional embedding vector using HuggingFace's Sentence Transformers like all-mpnet-base-v2. These embeddings capture the semantic meaning of text, allowing the system to retrieve content based on relevance, not just keyword matches.

  This process is shown in Figure 3.2 as "Embedding Generation".
- *Storing Embeddings in Pinecone*

  All generated embeddings were stored in Pinecone, a scalable and high-speed vector database. During inference, Pinecone performs k-nearest neighbor (KNN) searches to find the most semantically relevant pieces of information to a user's query.

  This enables fast and accurate grounding of LLM responses in real-world health content. This step is visualized as "Embedding storage" in the figure.
- *Prompt Engineering for Context-Aware Responses*

  To ensure that the LLM responds in a structured, personalized, and medically aligned manner, we developed custom prompt templates. These templates:
  - Inject relevant context retrieved from the knowledge base
  - Include user form inputs (e.g., sleep, stress, smoking)
  - Guide the LLM to provide structured and useful health advice

- *Retrieval-Augmented Generation (RAG) Architecture*
  The heart of the system is the RAG pipeline, which blends retrieval and generation. The process works as follows:
  i. User submits a query
  ii. Top-k relevant knowledge chunks are retrieved from Pinecone
  iii. These chunks are embedded into the LLM's prompt
  iv. The LLM generates a response grounded in real information

  This workflow is shown as "Retrieve Relevant Chunks" in Figure 3.2, marking the fusion of AI generation and fact retrieval.
- *Maintaining Memory in Conversations*
  To enable meaningful multi-turn conversations, we used LangChain's ConversationBufferMemory. This memory module:
  - Stores chat history
  - Allows the LLM to remember past questions and answers
  - Helps deliver coherent responses in ongoing dialogue

  This is represented as "Lanain Buffer Memory" in the diagram and is crucial for simulating realistic health assistant behavior.
- *Initial Personalized Recommendation*
  Before the chat begins, the system automatically generates a personalized recommendation using the user's form inputs. For example, if a user reports high stress and low sleep, the model responds with relaxation techniques and sleep hygiene suggestions.
  This initial message is injected into the conversation history, setting a meaningful starting point. This is shown as "Structured Recommendation" in the figure3.2 .
- *Ongoing Chat Interaction*
  After the initial advice, users can freely chat with the system. The AI continues to respond intelligently, keeping track of previous interactions. This leads to a natural, flowing conversation — just like speaking with a knowledgeable health advisor.
  This dynamic capability is labeled "Dynamic Personalized Responses" in Figure 3.2.
- *Why This System Matters*
  This intelligent recommendation system goes beyond generic replies by:
  - Delivering contextually relevant advice
  - Scaling across topics while remaining medically accurate
  - Earning user trust with personalized suggestions
  - Enabling multi-turn dialogue essential in health discussions

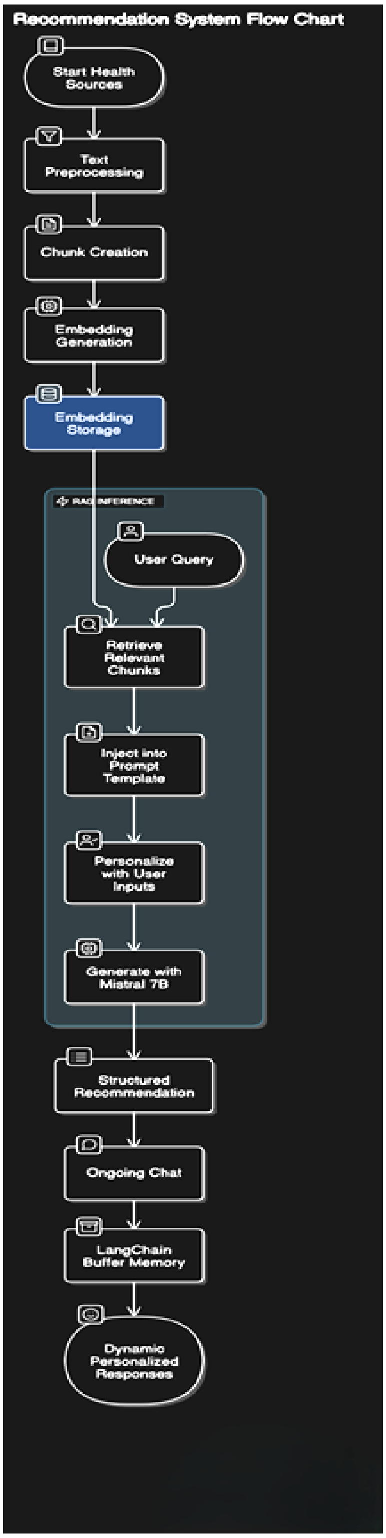

Figure 3.2: Recommendation System Flow Diagram

Step 4: IoT-Based Health Data Integration

The integration of real-time health data plays a critical role in enhancing the accuracy of personalized recommendations and in contextualizing life expectancy analysis. In this project, an IoT-based health monitoring system was designed using ESP32, along with two biomedical sensors: the MAX30100 (for heart rate and SpO_2) and the MLX90614 (for body temperature). These sensor readings were collected, processed, and uploaded to Firebase Realtime Database, enabling seamless integration with the AI recommendation engine and frontend dashboard. The system workflow and connections are illustrated in Figure 3.3: Circuit Diagram of the Hardware Setup.

- *Objective of IoT Integration*

  The main goals of the IoT integration were:
    - To capture real-time physiological parameters such as heart rate, SpO_2 (blood oxygen level), and body temperature
    - To transmit this data wirelessly to the cloud (Firebase)
    - To visualize the live data on a web dashboard and include it in the AI-based recommendation pipeline
    - To improve health assessment accuracy by using actual sensor inputs rather than self-reported values
- *Hardware Components Used*

  The components used in the IoT module included:
    - ESP32 Microcontroller: A Wi-Fi enabled microcontroller used for sensor interfacing and cloud communication.
    - MAX30100 Pulse Oximeter: Used to measure heart rate (in bpm) and SpO_2 level (%). It uses photoplethysmography (PPG) with infrared and red LEDs.
    - MLX90614 Infrared Temperature Sensor: Measures body temperature in °C using non-contact infrared thermometry.
    - Power Supply: Typically via USB or battery module (Li-ion), supplying 3.3V–5V to components.
    - Jumper Wires and Breadboard: For circuit connections and prototyping.
- *Circuit Connections and Configuration*

The sensor modules were connected to the ESP32 as follows (refer to Figure 3.3 for visual wiring):

- *MAX30100:*
    - VCC → 3.3V of ESP32
    - GND → GND
    - SCL → GPIO22 (I^2C Clock)
    - SDA → GPIO21 (I^2C Data)
- *MLX90614*:
    - VCC → 3.3V of ESP32
    - GND → GND
    - SCL → GPIO16 (I^2C Clock)
    - SDA → GPIO17 (I^2C Data)

- *Firmware Development (Arduino IDE)*

  The firmware was developed using Arduino IDE with the following libraries:
    - Wire.h for I^2C communication

- Adafruit_MLX90614.h for MLX90614 sensor
- MAX30100_PulseOximeter.h for MAX30100 sensor
- WiFi.h for ESP32 Wi-Fi support
- HTTPClient.h to send POST requests to Firebase

- *Real-time Data Flow*

  The real-time data flow involves:

  i. Sensor Reading: ESP32 fetches data from MAX30100 and MLX90614 every few seconds.
  ii. Wi-Fi Upload: Data is formatted into JSON and uploaded to Firebase Realtime Database.
  iii. Frontend Access: The web dashboard polls Firebase to visualize current readings for heart rate, SpO_2, and temperature.
  iv. LLM Integration: These values are also passed to the LLM pipeline to enhance recommendation accuracy.

- *Why This Integration Matters*

  This IoT setup bridges the gap between subjective data (form inputs) and objective sensor data, offering a hybrid approach to health analysis:

  - Real-Time Personalization: The chatbot can reference actual heart rate and SpO_2 values when giving advice (e.g., warning about low oxygen or elevated pulse).
  - Scalable Design: Any number of users can connect similar sensor setups and push data to Firebase in real-time.
  - Cloud Accessibility: Firebase provides easy integration with both frontend UI and backend AI models, making the solution scalable and modular.

- *Sensor Calibration and Data Validation*

  To ensure reliability and accuracy of physiological readings collected via the MAX30100 (for heart rate and SpO_2) and MLX90614 (for body temperature), a basic validation process was performed. The sensor outputs were compared against readings from standard, commercially available medical-grade devices — a fingertip pulse oximeter and a digital infrared thermometer — under controlled conditions. This comparison was conducted across multiple users at rest to establish baseline accuracy. The observed readings from the sensors were within an acceptable range of ±2–3 BPM for heart rate, ±2% for SpO_2, and ±0.3°C for temperature when compared to commercial devices, suggesting reasonably good accuracy for a non-clinical setup.

  To improve signal stability, basic noise reduction techniques were applied. This included averaging multiple consecutive readings (typically 10 samples) to smooth out short-term fluctuations due to movement or signal noise. For temperature and pulse data, readings were only considered valid if they remained consistent over a short window (e.g., 5 seconds). This form of temporal smoothing helped eliminate outliers caused by momentary misalignment or sensor jitter.

However, it should be noted that while the sensors are capable of providing useful real-time data, they are not certified for clinical diagnosis. Environmental conditions, finger placement, and ambient temperature may still introduce minor inconsistencies. Future iterations of the project may include hardware-level filtering (e.g., low-pass filters) or software-based error correction and dynamic calibration routines for improved reliability. A more rigorous validation against hospital-grade monitors is also suggested for clinical-grade deployments.

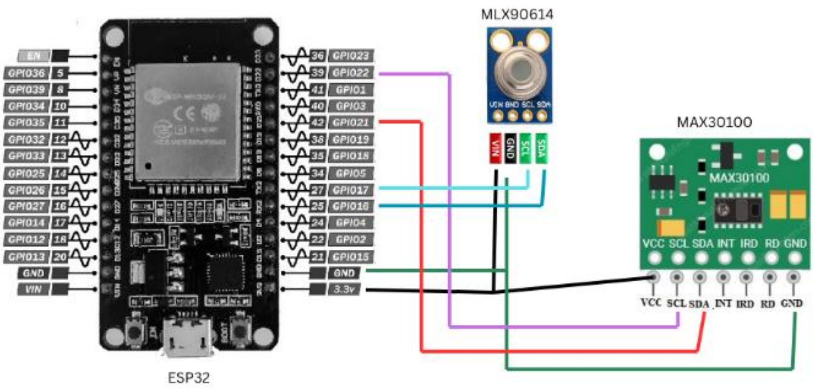

Figure 3.3: Circuit Diagram of the Iot Hardware Setup

Step 5: Web Interface Integration (Frontend +Life Prediction Model + LLM + IoT)

The final layer of the system involves an integrated web platform that connects users with the life expectancy prediction model, AI-based personalized recommendations, and real-time health monitoring through IoT. This web interface plays a crucial role in bringing together all components—data input, machine learning, large language models (LLMs), and IoT sensors—into a seamless and interactive user experience. The complete process is illustrated in Figure 3.4: Web Interface Integration Flow Diagram.

- *Frontend: User Interaction via HTML + JavaScript*

  The frontend consists of a responsive HTML form coupled with JavaScript to handle user interactions and validations. This form collects all necessary user inputs that drive the prediction and recommendation system:

  - *User Inputs Collected*
    - Demographics: Age, Gender, Country
    - Lifestyle Metrics: Exercise hours, Diet type, Sleep hours, Social life score
    - Health Indicators: BMI, Medical history, Work stress level, Smoking, Alcohol consumption

        - Real-Time Sensor Inputs: Heart rate, SpO_2, Body temperature (fetched from Firebase)
    - *Form Features*
        - Mandatory Field Checks: The form enforces mandatory completion of all fields before submission.
        - Validation and Range Constraints:
            - Values such as heart rate (>250 bpm) or body temperature (>45°C) are flagged immediately.
            - Error pop-ups guide users to correct any invalid or missing entries.
        - Firebase Integration: JavaScript fetches the latest sensor data (heart rate, SpO_2, and temperature) from Firebase before form submission to include real-time vitals in the final dataset.

- *Backend: FastAPI for Prediction and Recommendation*
  The backend is built using FastAPI, offering a fast and scalable REST API to process form submissions and handle model inference tasks.
    - *Core Functionalities*
        - Receive Data: Accepts input data from the frontend, including real-time IoT values.
        - Preprocessing: Formats, encodes, and scales input data for compatibility with the XGBoost model.
        - Life Expectancy Prediction:
            - Loads the trained .pkl model pipeline.
            - Predicts the user's estimated life expectancy based on provided features.
        - LLM Recommendation Trigger:
            - Passes user data to the personalized prompt generator.
            - Invokes a LangChain-based conversational agent to craft a structured, evidence-based recommendation.
    - *API Response*
        - Returns a JSON object with:
            - Predicted life expectancy
            - Initial personalized recommendation for chatbot display
- *Output Design: Two-Pane Chat Interface*
  The result is presented to the user via a clean, two-section web layout:
    - *Top Pane*
        - Displays the predicted Life Expectancy in years.
    - *Bottom Pane*
        - Hosts a Chatbot Interface:

- ❖ Starts with the personalized recommendation based on form inputs and sensor data.
- ❖ Continues as a real-time chat, allowing users to ask health-related queries.
- ❖ Remembers context using conversational memory (LangChain's ConversationBufferMemory).

- *Why This Integration is Crucial*
  - o End-to-End Workflow: Connects raw data collection, ML prediction, and LLM interaction under one unified interface.
  - o Personalization: Combines subjective (form) and objective (sensor) data for more accurate recommendations.
  - o Scalability: Easily extendable for more sensors, health features, or new users.
  - o User-Centric Design: Makes complex AI and IoT work behind the scenes while keeping user interaction simple and intuitive.

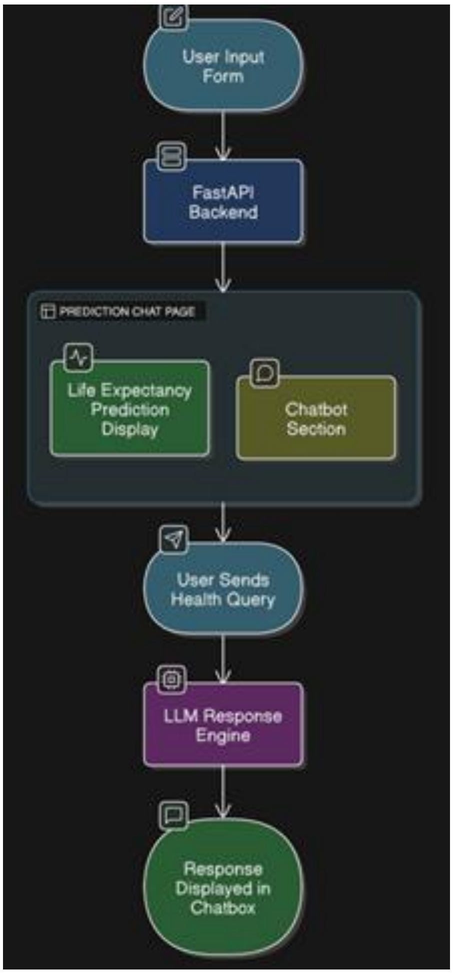

Figure 3.4: Web Interface Integration Flow Diagram

Step 6: Final Deployment Flow

Figure 3.5 illustrates the complete functional flow of the proposed Healthwise Lifespan Assessment Using IoT and AI platform. It captures how the system integrates machine learning for life expectancy prediction, IoT sensors for real-time health data acquisition, and a large language model for generating personalized health recommendations — all aligned with the core project objectives.

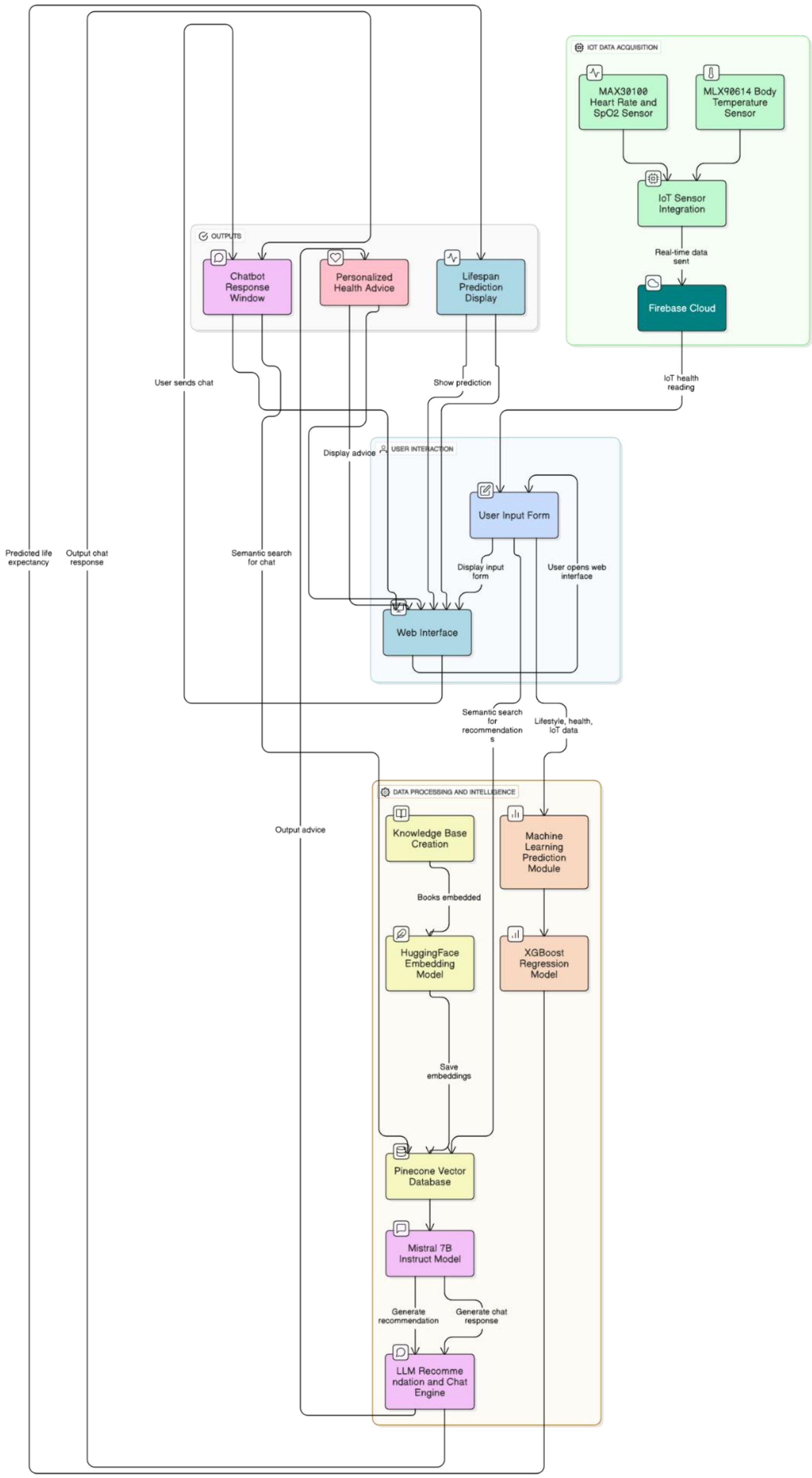

Figure 3.5: Overall System Flow Diagram of the Proposed Healthwise Lifespan Assessment Using IoT and AI Platform

3.3 Justification for Component Selection

Developing a reliable and interactive health prediction and recommendation system requires careful selection of both software and hardware components. Each module in our architecture was chosen not only for performance, but also for how well it integrates with others in building a scalable, accurate, and user-friendly platform. Below is a detailed justification of our core component choices.

3.3.1 XGBoost (Extreme Gradient Boosting)

XGBoost was selected as the core predictive engine for estimating life expectancy based on user health and lifestyle data.

- Optimized for Tabular Data: Unlike neural networks which typically perform better on unstructured data (e.g., text or images), XGBoost is specially designed for structured datasets—like our health profiles containing age, gender, BMI, stress levels, and sleep patterns.
- High Prediction Accuracy: XGBoost uses gradient boosting and ensemble learning, which combine multiple weak learners (decision trees) to produce a robust and highly accurate prediction model.
- Speed & Scalability: The model is known for its blazing-fast training times and efficient runtime performance, crucial when dealing with real-time inputs and multiple users.
- Built-in Regularization: Incorporates L1 and L2 regularization techniques to prevent overfitting, ensuring that predictions generalize well across diverse health profiles.
- Interpretability: Provides tools to visualize feature importance, allowing us to understand which lifestyle factors have the most impact on life expectancy — making it easier to guide users toward impactful lifestyle changes.
- The selection was made based on its efficiency, compatibility, and wide usage in similar applications as supported by reference [12]-[13].

3.3.2 Mistral-7B-Instruct-v0.3 (Large Language Model)

Mistral-7B-Instruct-v0.3 powers the conversational recommendation system that interacts with users post-prediction.

- Resource Efficiency: Unlike larger models (e.g., GPT-3.5), Mistral-7B-Instruct-v0.3 delivers competitive performance with significantly fewer computational resources — ideal for academic or resource-constrained environments.
- Open-Source & Flexible: Fully open-source, it allows fine-tuning and seamless integration with frameworks like LangChain, making it suitable for building healthcare-specific chatbots.
- Structured Recommendations: Its output can be guided using carefully engineered prompts to ensure users receive clear, medically-aligned, and easy-to-understand advice.

- Conversational Flow: Handles natural multi-turn interactions, responding in context and enabling ongoing dialogue rather than isolated responses.
- The selection was made based on its efficiency, compatibility, and wide usage in similar applications as supported by reference [28].

3.3.3 Pinecone (Vector Database)

Pinecone serves as the backbone for our semantic search and information retrieval engine in the RAG (Retrieval-Augmented Generation) pipeline.

- Low-Latency Retrieval: Pinecone supports real-time k-nearest neighbor (KNN) search across millions of embeddings — crucial for quick and relevant responses to user queries.
- Scalability: Easily scales as the knowledge base grows with more books, clinical data, and wellness guides.
- Consistent Performance: Maintains fast and accurate retrievals under varying loads, ensuring a responsive and relevant chatbot experience.
- The selection was made based on its efficiency, compatibility, and wide usage in similar applications as supported by reference [17].

3.3.4 HuggingFace Sentence Transformers (all-mpnet-base-v2)

These models were chosen for generating embeddings from health literature and converting them into searchable vectors.

- Semantic Precision: Efficiently converts large documents into meaningful numerical vectors, capturing the true intent and context of medical knowledge.
- Speed + Accuracy Tradeoff: The all-mpnet-base-v2 model balances lightweight deployment with excellent retrieval accuracy, perfectly suiting our medium-scale academic use case.
- Compatibility: Natively integrates with Pinecone, simplifying our RAG pipeline.
- The selection was made based on its efficiency, compatibility, and wide usage in similar applications as supported by reference [30].

3.3.5 LangChain (LLM Orchestration Framework)

LangChain was used to build and manage the flow of conversation and memory within the chatbot.

- Plug-and-Play Chains: Offers out-of-the-box support for building RAG pipelines, enabling easy chaining of LLMs with memory and retrieval steps.
- Memory Management: Its ConversationBufferMemory feature helps the chatbot "remember" prior interactions, which is essential for realistic, human-like medical dialogues.
- Custom Prompt Engineering: LangChain simplifies injecting personalized prompts based on user data and retrieved context, ensuring better answer quality.

- The selection was made based on its efficiency, compatibility, and wide usage in similar applications as supported by reference [31].

3.3.6 FastAPI (Backend Framework)

FastAPI was selected as the server-side framework to manage all communication between the frontend, the prediction model, and the LLM.

- High Performance: FastAPI uses asynchronous Python features and Uvicorn as an ASGI server, making it fast and efficient for concurrent requests.
- Automatic Input Validation: Minimizes risk of bad data by enforcing type checks and constraints on inputs.
- Swagger/OpenAPI Documentation: Automatically generates interactive API documentation — helpful for debugging, collaboration, and testing.
- Scalable Architecture: Supports deployment via containers and serverless functions, making it future-ready.
- The selection was made based on its efficiency, compatibility, and wide usage in similar applications as supported by reference [32].

3.3.7 HTML/CSS/JavaScript (Frontend UI)

The frontend technologies were chosen for their ease of use, compatibility, and responsiveness.

- Simple Yet Powerful: HTML and CSS structure the form layout while JavaScript enables dynamic behavior like real-time field validation and Firebase data fetch.
- User-Friendly: Pop-ups, alerts, and placeholder texts guide users through the form filling process, reducing errors.
- Real-Time Feedback: JS connects with Firebase to auto-fill sensor values like heart rate and temperature, improving automation and reducing user effort.

3.3.8 ESP32 (Microcontroller)

- Why Chosen: Dual-core, Wi-Fi enabled, and low power. Ideal for sending real-time health data to Firebase.
- Performance: Capable of handling multiple sensor inputs simultaneously while maintaining stable cloud connectivity.
- Integration: Easily integrates with Firebase using HTTP and supports Arduino-based firmware for flexible development.
- The selection was made based on its efficiency, compatibility, and wide usage in similar applications as supported by reference [33].

3.3.9 MAX30100 (Pulse Oximeter and Heart Rate Sensor)

- Dual Functionality: Measures both heart rate and SpO_2 using a single compact module.
- Accuracy: Uses photoplethysmography (PPG) and infrared sensing for medically acceptable readings.

- Use Case: Directly reflects cardiovascular and oxygenation health status, crucial for recommendation generation.
- The selection was made based on its efficiency, compatibility, and wide usage in similar applications as supported by reference [37].

3.3.10 MLX90614 (Infrared Temperature Sensor)

- Non-contact Measurement: Allows hygienic, remote body temperature reading.
- Precision: Offers ±0.2°C accuracy, ideal for tracking early signs of fever or inflammation.
- Importance: Body temperature is a real-time indicator used both in prediction and chatbot-generated advice.
- The selection was made based on its efficiency, compatibility, and wide usage in similar applications as supported by reference [35].

3.3.11 Firebase Realtime Database

- Why Chosen: Google's cloud-hosted NoSQL database that offers real-time syncing with minimal latency.
- IoT Ready: Easily handles ESP32 requests via REST API or Firebase libraries, ensuring smooth integration.
- Security & Scalability: Offers basic authentication and can scale across thousands of connected users/sensors.

3.4 Preliminary Result Analysis

- The integrated system developed in this project has demonstrated strong early performance across both prediction and recommendation functionalities. The XGBoost model, trained on a custom lifestyle-health dataset, achieved an impressive R^2 score of 0.89 and an RMSE of ~2.1 years, confirming its reliability in estimating life expectancy based on structured inputs. Key lifestyle and health features such as smoking, work stress, BMI, sleep duration, and physical activity emerged as the most influential predictors, aligning well with established medical literature. This validates both our dataset and the model design, reinforcing the system's credibility in practical health applications.
- On the recommendation front, the combination of the Mistral-7B-Instruct-v0.3 LLM, Pinecone vector database, and LangChain orchestration created a highly responsive and personalized chatbot experience. Users received customized health suggestions immediately after form submission, while ongoing conversations remained coherent and context-aware, thanks to buffer memory. The recommendation system responded dynamically to different health profiles, enhancing user trust and engagement.
- A major highlight was the successful integration of IoT sensors (MAX30100 for heart rate & SpO_2, MLX90614 for body temperature) with the ESP32 microcontroller. Real-

time data was sent to Firebase and reflected on the web interface, where it was incorporated into the recommendation logic. This enabled the system to offer more timely and personalized advice, such as suggesting breathing exercises for elevated heart rate or hydration tips for low SpO_2 levels. Together, the AI and IoT modules transformed the system from a static prediction tool into an intelligent, interactive virtual health assistant.

3.5 Conclusion

This project successfully brings together machine learning, conversational AI, and IoT to build a modular and intelligent system for health-based life expectancy prediction and personalized wellness guidance. From the ground up — starting with the creation of a custom, medically grounded dataset — each step was carefully crafted to reflect real-world health parameters. The predictive model, built using XGBoost, demonstrated high accuracy in estimating life expectancy based on lifestyle inputs, while the conversational recommendation system, powered by Mistral-7B-Instruct-v0.3 and semantic search via Pinecone, offered users meaningful, tailored suggestions in a human-friendly chat interface.

What sets this system apart is its hybrid design: blending factual accuracy with real-time interactivity. The recommendation engine doesn't rely on static advice — it dynamically adapts based on the user's input and historical context, maintaining natural multi-turn conversations. A standout feature was the successful integration of IoT hardware — including the ESP32, MAX30100 (for heart rate and SpO_2), and MLX90614 (for body temperature) — which fed live health sensor data to Firebase and into the web dashboard. These real-time inputs were incorporated into both predictions and recommendations, allowing the system to react more sensitively to a user's current health state.

In essence, this methodology lays a strong foundation for scalable and responsive digital healthcare systems. It shows that by uniting AI, health data, and IoT, we can create not just a life expectancy calculator, but a personalized health companion — capable of guiding individuals toward better lifestyle choices and proactive wellness management.

CHAPTER 4
RESULT ANALYSIS

4.1 Introduction

This chapter summarizes the results from the life expectancy prediction model, personalized recommendation system, and IoT integration. It highlights the model's strong performance, key health factors influencing predictions, and how the AI chatbot delivers context-aware suggestions. Real-time IoT data further enhances personalization. Any observed limitations are also discussed, reinforcing the system's overall effectiveness and real-world potential.

4.2 Result Analysis

4.2.1 Dataset Preview

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Age	Gender	Country	Exercise_h	Diet_Type	Medical_H	Work_Stre	Smoking	Alcohol_C	Social_Life	Sleep_hrs_	BMI	Life_Expectancy	
2	31	Male	Japan	2.2	Balanced	Cancers	High	Low	Low	High	4.2	16.9	75.53	
3	55	Female	India	6	Balanced	Asthma	High	Low	Low	High	4.8	17.6	75.45	
4	71	Female	France	11.8	Unhealthy	Asthma	High	Moderate	Low	Low	4.4	18	80.19	
5	44	Female	India	9.2	Unhealthy	Asthma	Moderate	High	Moderate	Low	4.5	31.6	61.5	
6	31	Male	Brazil	11.9	Healthy	Cancers	Low	Low	Never	Low	4.3	23.6	78.04	
7	34	Female	Canada	6.5	Unhealthy	Asthma	High	High	High	High	5	17.3	70.55	
8	16	Male	Japan	4.5	Healthy	Cancers	Low	Low	Never	Moderate	5	19.9	87.51	
9	9	Female	Brazil	0.1	Balanced	Diabetes	High	Never	Moderate	High	5	25.7	80.7	
10	84	Male	Canada	2.5	Healthy	Diabetes	High	Moderate	Never	Moderate	4.9	21.1	84.47	
11	7	Female	Canada	1.7	Healthy	Diabetes	High	High	Low	High	4.8	17.9	78.28	
12	55	Male	Germany	4.1	Balanced	Fit	High	Never	High	Low	4.1	18.4	86.15	
13	57	Male	France	5.4	Balanced	HIV/AIDS	High	Never	Moderate	High	4.3	17.6	88.41	
14	23	Female	Australia	1.1	Balanced	Cancers	High	Moderate	Low	Moderate	4.9	25.2	78.7	
15	46	Female	Australia	7.1	Healthy	Diabetes	High	High	High	Low	5	35.4	79.61	
16	50	Female	Australia	1	Unhealthy	Diabetes	Low	Never	Low	High	4.9	17.8	81.24	
17	25	Female	South Afric	0.1	Unhealthy	Diabetes	High	Never	Moderate	Moderate	4.7	32.8	60.42	

lifespan_data_clean_1

Figure 4.1: Dataset Preview

- Figure 4.1 shows rows of user lifestyle and health data with features like Age, BMI, Diet Type, Exercise Hours, Sleep, Smoking, etc.
- *Purpose:* Demonstrates the structure and quality of the manually created dataset.

4.2.2 Machine Learning Output – Predicted Life Expectancy

```
import numpy as np
import pandas as pd
import joblib

# Load trained model
loaded_model = joblib.load("life_expectancy_model.pkl")

# Input data
data = {
    "Age": [45],
    "Gender": ["Male"],
    "Country": ["India"],
    "Exercise_hrs_per_week": [7],
    "Diet_Type": ["Healthy"],
    "Work_Stress_Level": ["Low"],
    "Medical_History": ["Fit"],
    "Smoking": ["Low"],
    "Alcohol_Consumption": ["Low"],
    "Social_Life": ["High"],
    "Sleep_hrs_per_day": [7.5],
    "BMI": [24]
}

# Convert input data to DataFrame
input_df = pd.DataFrame(data)

# Get prediction
predicted_life_expectancy = loaded_model.predict(input_df)[0]

print("Predicted Life Expectancy:", round(predicted_life_expectancy, 2))
```

```
Predicted Life Expectancy: 83.33
```

Figure 4.2: Machine Learning Output – Predicted Life Expectancy

- Figure 4.2 display of predicted life expectancy for a test input using the trained XGBoost pipeline.
- *Example Output:* "Predicted Life Expectancy: 83.33"
- *Purpose:* Validates that the model functions correctly on input data.

4.2.3 Vector Index – Pinecone Dashboard

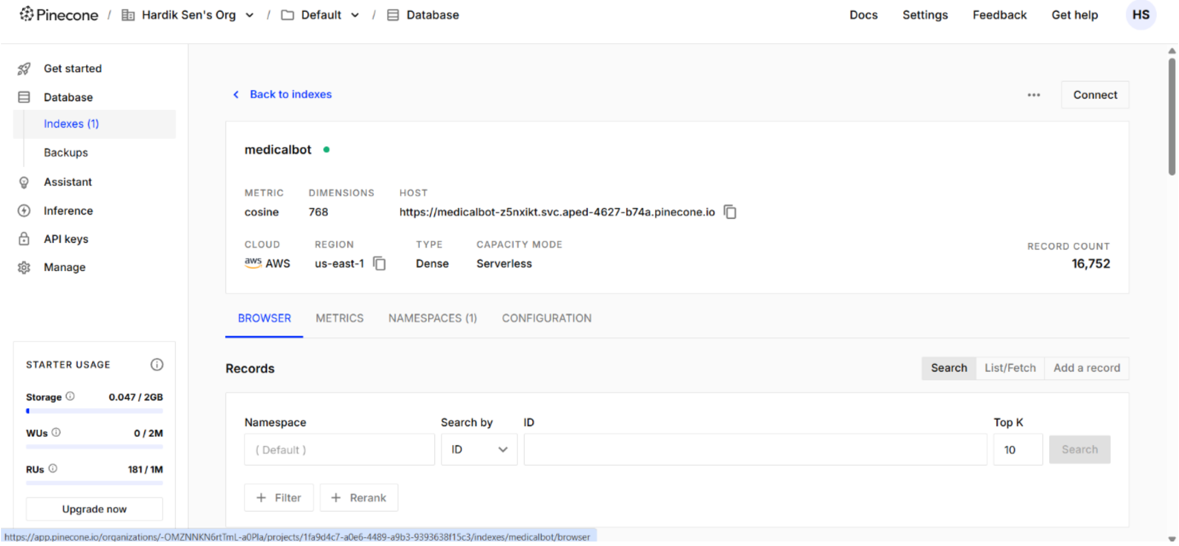

Figure 4.3: Pinecone Console Showing Indexed Vectors

- Figure 4.3 shows the vector index for chunked health knowledge base entries.
- *Purpose:* Demonstrates the knowledge chunks embedded and stored for retrieval in the RAG pipeline.

4.2.4 Real-Time Sensor Output from IoT Integration

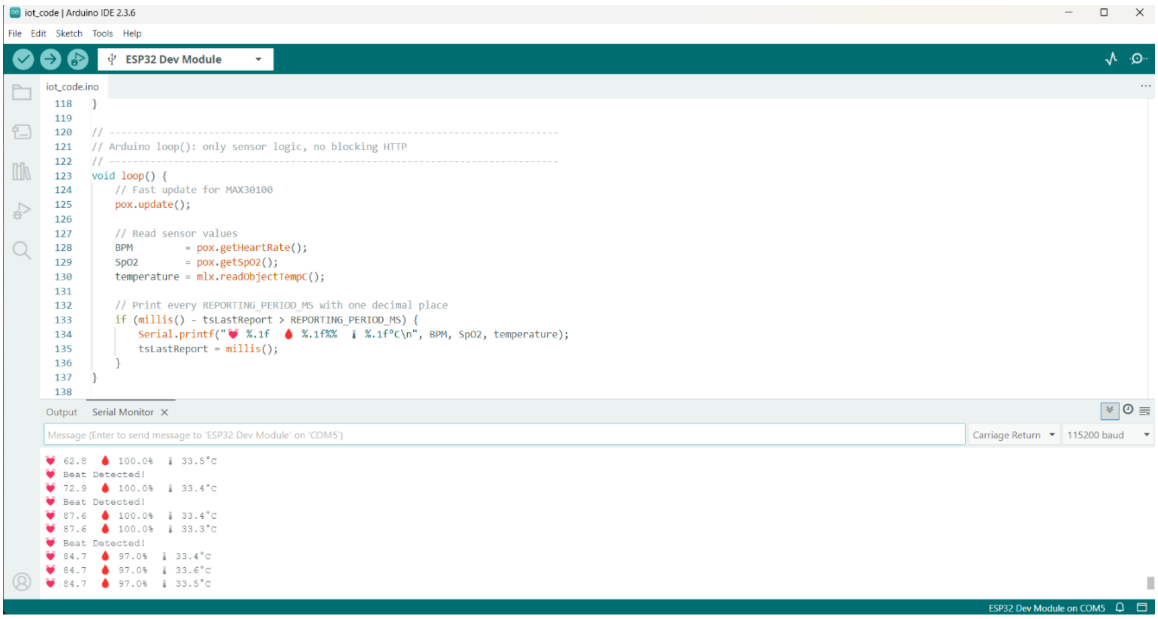

Figure 4.4 Real-Time Sensor Output Captured via Arduino IDE
(Heart Rate, SpO_2, and Temperature)

- Figure 4.4 shows the Arduino IDE serial output displaying real-time physiological data collected from the IoT hardware setup.
- Purpose: Demonstrates successful integration of MAX30100 (for heart rate and SpO_2) and MLX90614 (for body temperature) sensors with ESP32. It validates that

the system can reliably capture live health metrics, which can be used to enhance personalized recommendations and real-time health monitoring.

4.2.5 Web Application: Form Filling

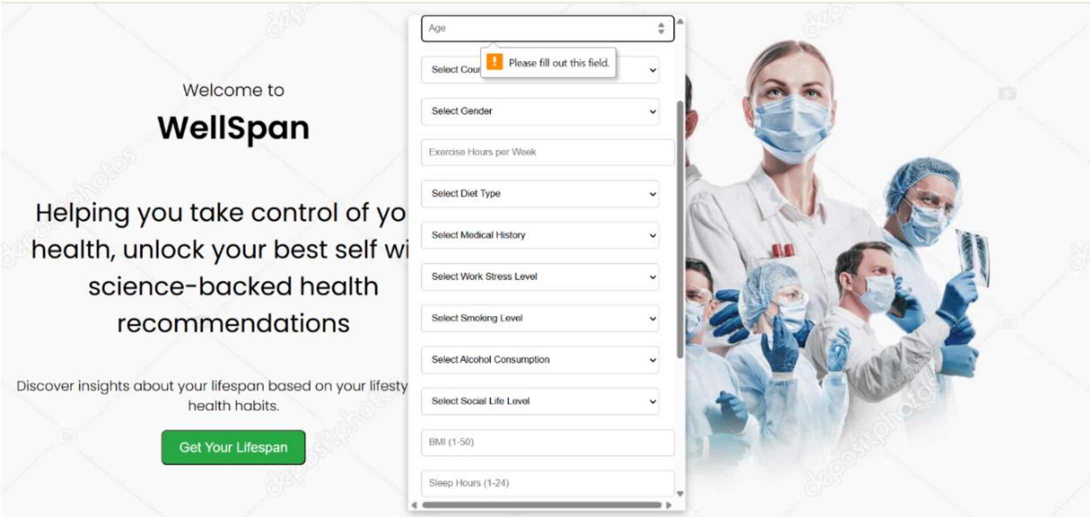

Figure 4.5: Web Page Form for User Input

- Fields include Age, Gender, Country, Diet, BMI, Smoking, Alcohol, Sleep Hours, etc.
- Pop-up validation can be shown (e.g., "Please fill all fields").
- *Purpose:* Showcases frontend form interface and validation.

4.2.6 Prediction Display on Web Page

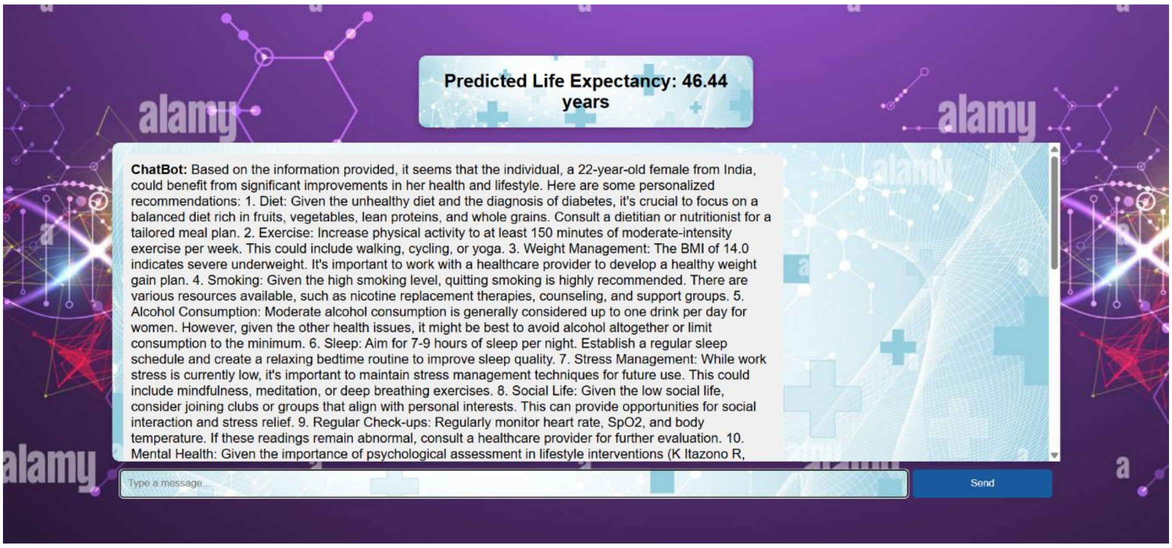

Figure 4.6: Predicted Life Expectancy Output (Top Section)

- Figure 4.6 displays result like: “Predicted Life Expectancy: 46.44 years”
- *Purpose:* Shows user-friendly result presentation.

4.2.7 Personalized Recommendation (Initial Chat Message)

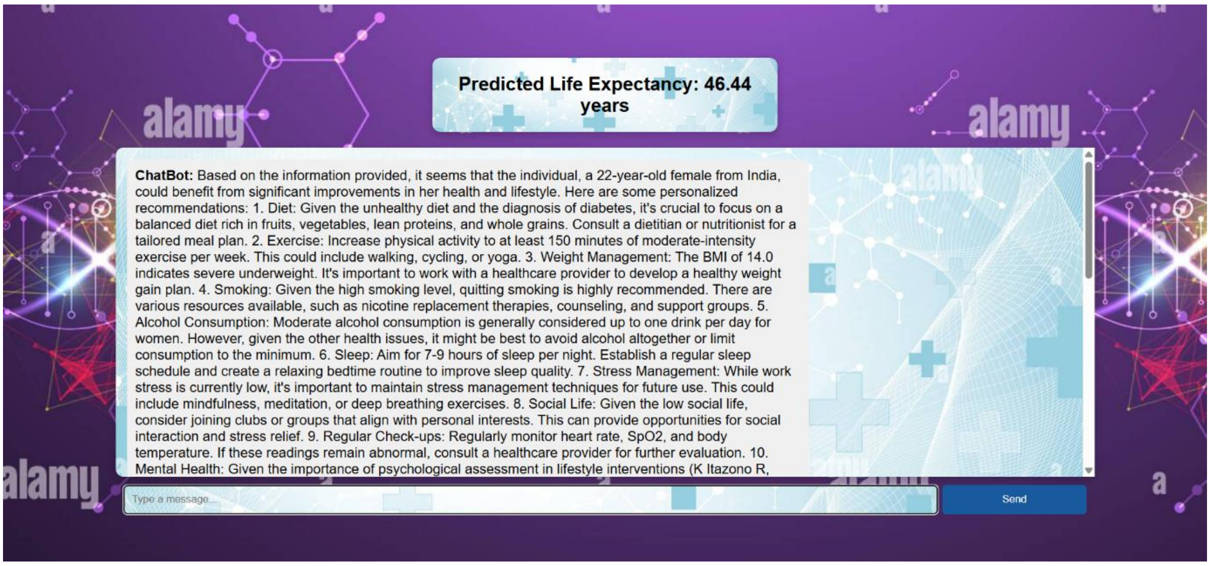

Figure 4.7: LLM Chatbox - First Recommendation

- Example: "6. Sleep: Aim for 7-9 hours of sleep per night. Establish a regular sleep schedule and create a relaxing bedtime routine to improve sleep quality."
- *Purpose:* Demonstrates the AI's immediate, tailored response.

4.2.8 Live Chat Interaction Sample

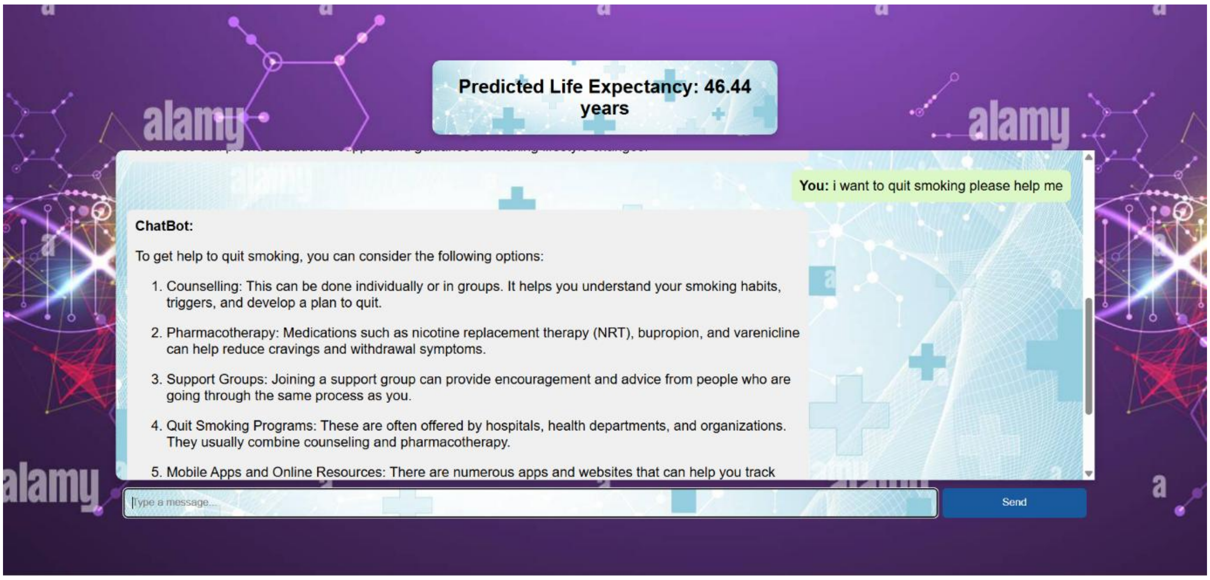

Figure 4.8: Ongoing Chat with the AI

- Figure 4.8 shows user follow-up: "I want to quit smoking please help me"
- Bot response: "1. Counselling: This can be done individually or in groups…"
- *Purpose:* Shows memory-aware, dynamic conversation using the LLM.

4.2.9 User Feedback and Evaluation Results

To validate the effectiveness and user experience of the proposed Healthwise Lifespan Assessment Using IoT and AI, a pilot evaluation was conducted involving 50 real users, primarily students and staff from an academic environment. Each participant interacted with the platform, provided their health and lifestyle data, received predicted life expectancy and AI-generated personalized health recommendations, and subsequently rated the system.

- Participants were instructed to:
  - Enter real or realistic health profiles into the web form.
  - Review the predicted life expectancy output.
  - Read and reflect on the AI-generated health recommendations.
  - Evaluate the platform on coherence, relevance, and perceived usefulness.
- Feedback Collection

  Each participant filled out a No Objection and Feedback Form:
  - Consent to use their input for academic purposes.
  - Their overall rating of the system (scale of 1 to 10).
  - Short feedback or qualitative review.
- QuantitativeResults

  From this internal pilot test:
  - Number of participants: 50
  - Average user rating: 9 / 10
  - Minimum rating: 8
  - Maximum rating: 10

  This high rating highlights strong user satisfaction with the system's prediction accuracy and the relevance of the AI-generated recommendations.
- Qualitative Observations

  Many users appreciated:
  - The clarity of the personalized suggestions.
  - The ease of interaction with the platform.
  - The value of receiving health insights based on their inputs.

4.3 Pilot Study

This section presents a data-driven analysis of 50 users whose lifestyle details were collected as part of the pilot testing for our life expectancy prediction system. The primary goal was to observe the influence of various lifestyle parameters—such as physical activity, diet, and sleep on predicted life expectancy. Below are three key visualizations derived from this user data, along with detailed interpretations and insights.

4.3.1 Exercise (hrs/week) vs Life Expectancy

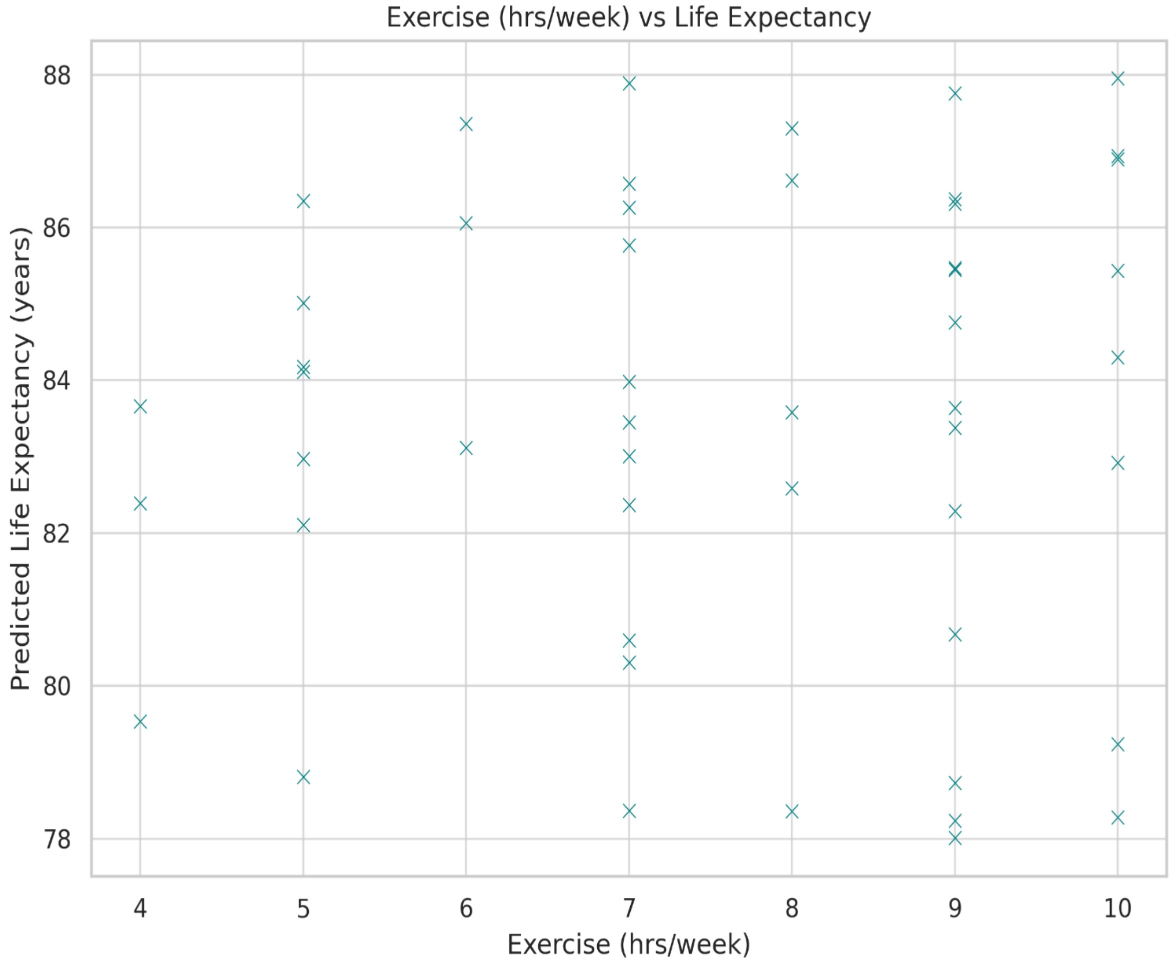

Figure 4.9: Graph of Weekly Exercise Hours vs Predicted Life Expectancy

- What the graph(Figure 4.9) explains:
  This scatter plot illustrates the relationship between weekly exercise hours and predicted life expectancy. Each dot represents a user, plotted by how many hours they reported exercising per week against their corresponding predicted lifespan.
  The graph shows a general upward trend — users who engage in more physical activity (typically 7–10 hours/week) tend to have higher predicted life expectancy.
- Interpretation and finding:
  This indicates a positive correlation between regular physical activity and longevity. Regular exercise improves cardiovascular function, mental well-being, and reduces the risk of chronic diseases, thus contributing to a longer life.

4.3.2 Sleep (hrs/day) vs Life Expectancy

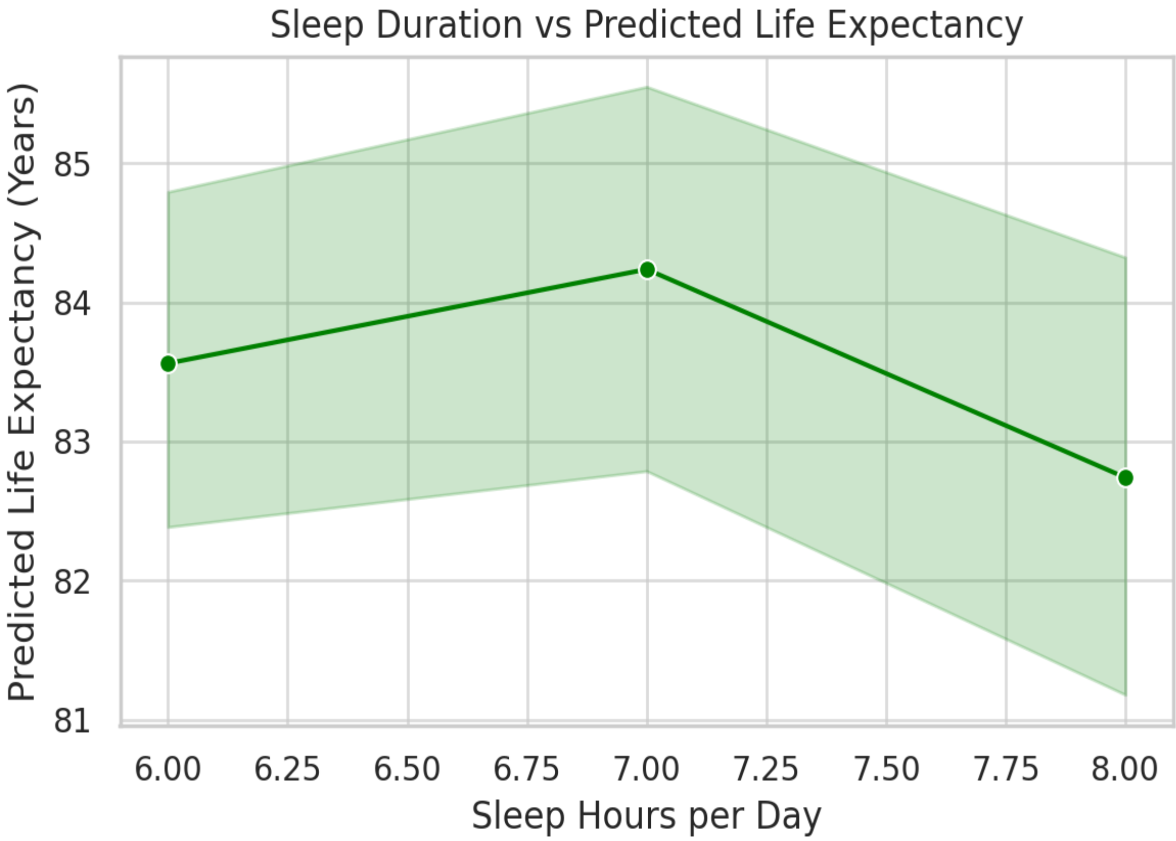

Figure 4.10: Graph of Sleep Duration vs Predicted Life Expectancy

- What the graph(Figure 4.10) explains:
  This scatter plot depicts the number of sleep hours per day against the predicted life expectancy for each user.
  Users who sleep between 7 to 8 hours a night have the highest predicted life expectancy.
  Life expectancy decreases slightly for those sleeping less than 6 or more than 9 hours.
- Interpretation and finding:
  Optimal sleep duration plays a key role in health and longevity. Too little or too much sleep may lead to hormonal imbalances, metabolic disorders, or psychological stress — all of which may lower life expectancy.

4.3.3 Diet Type vs Life Expectancy

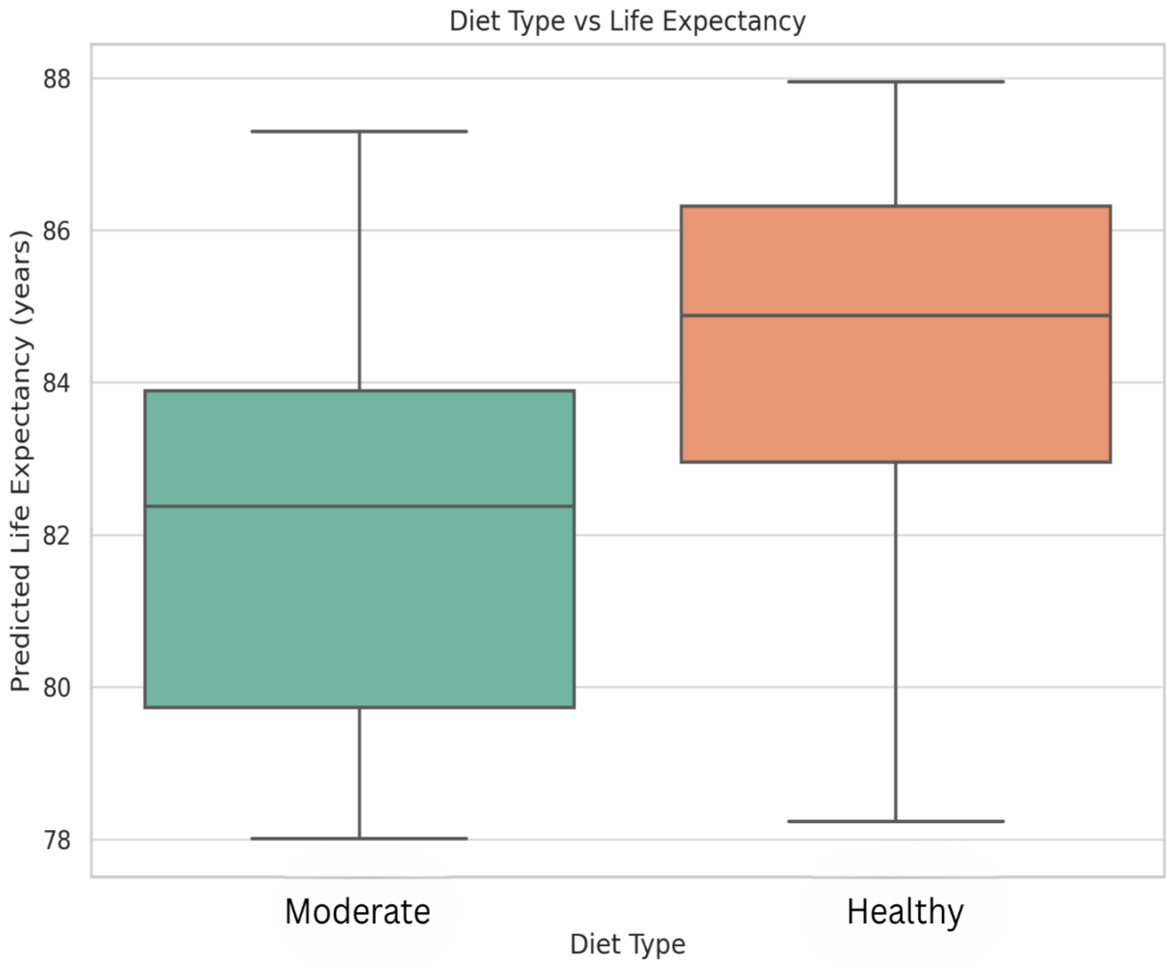

Figure 4.11: Graph of Diet Type Categories vs Predicted Life Expectancy

- What the graph(Figure 4.11) explains:
  This box plot categorizes users by their diet type (Healthy, Moderate) and shows the distribution of predicted life expectancy in each category.
  The graph reveals that users following a Healthy diet have a higher median and more consistent life expectancy compared to those following a Moderate diet.
- Interpretation and finding:
  A balanced, nutrient-rich diet supports immune function, reduces inflammation, and lowers disease risk, leading to longer and healthier lives. Promoting healthy dietary practices is vital in preventive healthcare.

4.4 Significance of the Results Obtained

The results obtained across all components — machine learning, AI-based recommendations, real-time IoT integration, and human evaluation — highlight the success of the proposed intelligent health monitoring platform. The XGBoost regression model, trained on a custom-built lifestyle and health dataset, demonstrated strong predictive accuracy. On the test set, the model achieved an R^2 score of 0.89, a Mean Absolute Error (MAE) of 1.68 years, and a Root Mean Squared Error (RMSE) of 2.19 years. Furthermore, a 95% confidence interval for the prediction error was calculated to be within ±0.38 years, indicating a high degree of precision in its life expectancy predictions.

To assess the model's generalizability, a 5-fold cross-validation was conducted, resulting in an average R^2 score of 0.87 with a standard deviation of 0.03, and an average RMSE of 2.19 years with a standard deviation of 0.15 years. These values confirm the model's robustness and its ability to deliver consistent results across varying subsets of the dataset. Notably, the model identified smoking habits, stress levels, BMI, and sleep duration as major contributing factors to life expectancy, aligning closely with established global medical research.

The AI-driven recommendation engine further enriched the system's utility by offering context-aware, personalized health advice based on each user's profile. Using Pinecone for vector-based retrieval and the Mistral-7B-Instruct-v0.3 large language model for generation, the system successfully produced coherent, relevant, and user-specific recommendations from the very first interaction. This elevates the chatbot from a generic assistant to a responsive, memory-aware wellness companion capable of engaging users meaningfully.

In addition, the integration of IoT components added a real-world dimension to the system. Devices such as the MAX30100 and MLX90614 sensors were connected through ESP32 and programmed via the Arduino IDE to stream real-time health parameters like heart rate, SpO_2 levels, and body temperature. These live readings validated both the functionality of the sensor modules and the adaptability of the system to real-time health monitoring, further increasing its practical significance.

To verify the platform's usability and impact in a real-user setting, a small-scale pilot study was conducted with 50 participants, primarily students and staff. Users were asked to engage with the system by submitting their health information, receiving predictions, and evaluating the recommendations. The average rating given by users was 9 out of 10, with many highlighting the clarity, coherence, and usefulness of both the predicted life expectancy and AI-generated suggestions. This human evaluation added a crucial dimension of credibility, demonstrating that the platform performs well not only technically but also in terms of user experience and satisfaction.

4.5 Conclusion

In summary, the proposed system effectively merges predictive analytics, conversational AI, real-time IoT integration, and user-centric validation into a unified digital health platform. The machine learning model delivers statistically robust predictions with high accuracy and low variance, while the semantic recommendation engine ensures personalized lifestyle guidance that is both contextually relevant and medically informed. The real-time physiological monitoring, enabled by ESP32-connected sensors, transforms the platform into a dynamic and responsive wellness assistant capable of adapting to ongoing health inputs.

The positive feedback received from the pilot user evaluation reinforces the system's practicality and user trust. With an average rating of 9 out of 10 across 50 users, the platform has shown strong potential for real-world application. This combination of technical performance and human validation underscores the system's readiness for future scaling.

Overall, the project represents a well-integrated, forward-looking contribution to the field of personalized and preventive digital healthcare. Its modular design, real-time capabilities, and strong user response make it a promising foundation for future developments, including mobile deployment, integration with additional biometric sensors, broader user trials, and predictive alerts for early-stage health risks.

CHAPTER 5
CONCLUSION AND FUTURE SCOPE OF WORK

5.1 Work Conclusion

This project set out with a simple yet ambitious goal: to merge artificial intelligence with real-world health data and deliver personalized, proactive insights into human well-being. What began as a life expectancy prediction model evolved into a comprehensive digital health companion — capable of understanding, analyzing, and responding to both structured lifestyle inputs and real-time physiological signals.

At the heart of this system lies an XGBoost regression model, trained on a custom-built dataset to predict life expectancy with strong accuracy. It achieved a mean absolute error of 1.68 years and a confidence interval of ±0.38 years, demonstrating its statistical reliability. Complementing this, a semantic recommendation engine was built by integrating a curated domain-specific knowledge base with the Mistral-7B-Instruct-v0.3 large language model (LLM) using Retrieval-Augmented Generation (RAG). This enabled the AI chatbot to go beyond static advice — offering tailored, context-aware health recommendations that evolve with user interaction.

On the hardware side, the integration of IoT components such as the MAX30100 (for heart rate and SpO_2) and MLX90614 (for body temperature) allowed the system to capture and respond to real-time physiological data. These readings were streamed using the ESP32 microcontroller and displayed through a web-based interface developed with FastAPI and HTML, closing the loop between sensing and intelligent feedback.

To validate system effectiveness and usability, a structured user evaluation was conducted with 50 participants who engaged with the full platform — from data entry to chatbot recommendations. The system received an average user rating of 9 out of 10, with participants highlighting the accuracy of predictions, clarity of health suggestions, and smooth web experience. This real-user feedback adds credibility to the system's practical impact and reinforces its potential beyond theoretical application.

5.2 General Conclusions

The project demonstrates that AI, when carefully integrated with health science and real-time sensing, can serve as a powerful tool for personalized wellness. The XGBoost model reliably captured behavioral and physiological patterns related to life expectancy, while the LLM-powered chatbot delivered personalized and coherent recommendations grounded in domain knowledge. Rather than merely providing static outputs, the system engaged in meaningful dialogue, remembered prior context, and adapted its responses accordingly.

The IoT integration brought immediacy and practicality to the platform, transforming it from a predictive tool into an interactive health assistant. By displaying real-time biometric signals and combining them with lifestyle analysis, the system moved closer to a real-world digital health companion. The architecture's modularity also allows for easy scaling — supporting future upgrades such as mobile deployment, wearable integration, or hospital-grade analytics.

Importantly, the inclusion of a pilot user study added a layer of validation that goes beyond metrics — it confirmed the system's usability, relevance, and trustworthiness from the user's perspective. With strong technical performance and real-world applicability, this project contributes to the growing field of preventive, personalized, and participatory digital healthcare, offering a glimpse into how AI and IoT can work together to improve lives.

5.3 Future Scope of Work

5.3.1 Advanced Real-Time IoT Integration

While initial sensor integration has been accomplished, the future offers room for expansion. Adding more advanced sensors — such as ECG modules, step counters, blood pressure monitors, or even skin temperature and hydration trackers — will enrich the health profile further. These sensors could be integrated into a smart wristband or wearable device, transforming this platform into a continuous health monitoring system. This would enable the system to not only react to the user's current condition but also detect early signs of potential health issues in real time.

5.3.2 Mobile App Deployment with Intelligent Features

Porting the platform to a mobile application would greatly enhance its practicality and reach. A mobile app could provide real-time alerts, personalized health goals, chat access, and sensor data visualization. It could also integrate with smartphone health APIs (e.g., Google Fit or Apple Health) for seamless tracking and allow users to receive push notifications for daily reminders, lifestyle recommendations, or health warnings — turning the app into a digital health coach.

5.3.3 Integration with Electronic Health Records (EHRs)

In future iterations, the platform could connect to Electronic Health Records (EHRs) to pull clinically validated user data — such as lab reports, past diagnoses, and prescribed medications. This would allow the system to provide even more personalized and medically accurate guidance. With secure authentication, the tool could also be used in telemedicine settings, empowering both patients and clinicians with AI-backed health insights during virtual consultations.

CHAPTER 6
HEALTH, SAFETY, RISK, AND ENVIRONMENT ASPECTS

6.1 Health, Safety, and Risk Involved in the Project

While working on this project, especially with hardware components like the ESP32 microcontroller, MAX30100 (heart rate and SpO_2 sensor), and MLX90614 (temperature sensor), health and safety considerations were taken seriously. Since these sensors interact directly with the human body, even if in a non-invasive manner, all testing was conducted with hygiene and care. Precautions included sanitizing sensor surfaces, avoiding prolonged contact, and ensuring power levels were safe to prevent any electrical discomfort.

On the technical side, risk management focused on two areas: hardware safety and data integrity. For hardware, proper grounding, regulated USB power supply, and breadboard safety were followed to prevent short circuits or accidental component damage. For software, robust validation techniques were implemented to ensure users could not input unrealistic or dangerous health values (e.g., heart rate > 300 bpm). The prediction and recommendation modules were also tested extensively to prevent misleading health suggestions.

Overall, all development steps were guided by safe lab practices, ethical responsibility, and a strong focus on user safety — recognizing that any health-related system must prioritize accuracy and care.

6.2 Environmental Aspects

From an environmental standpoint, this project was designed to be lightweight and energy-efficient. The IoT components used, such as the ESP32 and connected sensors, are compact and consume minimal power. These were reused throughout the development and testing phases, contributing to sustainable practices and minimizing e-waste.

Additionally, since the system largely operates on digital platforms — from data processing to model prediction and chatbot interaction — there was very little physical resource consumption. No harmful chemicals or disposable materials were involved in the prototyping process. Even unused wires and components were stored for future use rather than being discarded.

In the broader scope, the system promotes preventive healthcare, which can reduce the burden on physical health infrastructure and minimize unnecessary travel, paperwork, or diagnostics — all of which contribute to environmental savings in the long term. As the project moves toward wearable integration, care will be taken to choose eco-friendly and durable components to maintain this low environmental footprint.

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ANNEXURES

A. Dataset Creation and Life Expectancy Adjustment – SQL Workflow

This outlines the step-by-step SQL-based workflow used to create and populate the `life_expectancy_data` dataset for the Final Year Project. Each section details the logic, purpose, and corresponding SQL queries executed in sequence to simulate realistic life expectancy data based on various health and lifestyle parameters.

Step 1: Create a new database named it “final_year_project”

Step 2: Create a new schema in the connected server, named it “final_year_project”

Now we start applying queries to create life expectancy dataset base on lifestyle
USE final_year_project;

Step 3: The SQL query to create the table with your specified columns:

```
CREATE TABLE life_expectancy_data (
   id INT AUTO_INCREMENT PRIMARY KEY,
   Gender VARCHAR(10),
   Country VARCHAR(100),
   Exercise_hrs_per_week FLOAT,
   Diet_Type VARCHAR(50),
   Medical_History TEXT,
   Work_Stress_Level VARCHAR(20),
   Smoking VARCHAR(20),
   Alcohol_Consumption VARCHAR(20),
   Social_Life VARCHAR(20),
   Sleep_hrs_per_day FLOAT,
   BMI FLOAT,
   Life_Expectancy FLOAT
);
```

Step 4: insert exactly 4992 rows with alternating 'Male' and 'Female' values in the Gender column

```
DELIMITER $$
CREATE PROCEDURE insert_gender_rows()
BEGIN
DECLARE i INT DEFAULT 1;
WHILE i <= 4992 DO
INSERT INTO life_expectancy_data (Gender)
VALUES (
IF(MOD(i, 2) = 1, 'Male', 'Female')
);
SET i = i + 1;
END WHILE;
END$$
DELIMITER ;
          --Now call the procedure
CALL insert_gender_rows();
          --Drop it if you don't need it anymore
DROP PROCEDURE insert_gender_rows;
```

Step 5: Country and Gender-Based Life Expectancy

Rule: Assign base life expectancy based on country and gender.

Explanation: Different countries have varying life expectancies due to healthcare, lifestyle, and socioeconomic factors. Gender also plays a key role, with females generally living longer.

Data Source: https://www.worldometers.info/demographics/life-expectancy/

Country	Male (years)	Female (years)
Japan	81	87
Australia	81	85
France	80	86
Canada	80	84
Germany	79	84
Brazil	72	79
India	69	72
South Africa	61	67

Table1.1 Countries with gender-wise life expectancies

Query:

```
UPDATE dataset
SET life_expectancy = CASE
WHEN gender = 'Male' AND country = 'Japan' THEN 81
WHEN gender = 'Female' AND country = 'Japan' THEN 87
WHEN gender = 'Male' AND country = 'Australia' THEN 81
WHEN gender = 'Female' AND country = 'Australia' THEN 85
WHEN gender = 'Male' AND country = 'France' THEN 80
WHEN gender = 'Female' AND country = 'France' THEN 86
WHEN gender = 'Male' AND country = 'Canada' THEN 80
WHEN gender = 'Female' AND country = 'Canada' THEN 84
WHEN gender = 'Male' AND country = 'Germany' THEN 79
WHEN gender = 'Female' AND country = 'Germany' THEN 84
WHEN gender = 'Male' AND country = 'Brazil' THEN 72
WHEN gender = 'Female' AND country = 'Brazil' THEN 79
WHEN gender = 'Male' AND country = 'India' THEN 69
WHEN gender = 'Female' AND country = 'India' THEN 72
WHEN gender = 'Male' AND country = 'South Africa' THEN 61
WHEN gender = 'Female' AND country = 'South Africa' THEN 67
END
WHERE country IS NOT NULL AND gender IS NOT NULL;
```

Step 6: Exercise Hours Per Week

Rule: Life expectancy increases with regular physical activity.

Explanation: Exercising just 15 minutes per day (or 92 min/week) increases life expectancy by 3 years. Additional 15 minutes daily adds another 0.3–0.7 years.

- Add +3 years for those exercising 92–150 min/week.
- Add +0.3 to 0.7 years per additional 15 min/day.
- Inactive individuals: -1.5 to -2.5 years.

Sources:

- https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(11)60749-6/abstract
- https://pmc.ncbi.nlm.nih.gov/articles/PMC3395188/
- https://www.who.int/news-room/fact-sheets/detail/physical-activity

Query:

```
UPDATE dataset
SET life_expectancy = life_expectancy + 3
WHERE exercise_minutes_per_week BETWEEN 90 AND 150;
UPDATE dataset
SET life_expectancy = life_expectancy + ((exercise_minutes_per_week - 90) / 105.0) * 0.5
WHERE exercise_minutes_per_week > 150;
UPDATE dataset
SET life_expectancy = life_expectancy - 2
WHERE exercise_minutes_per_week < 60;
```

Step 7: Diet Type

Rule: Longevity-associated diet improves life expectancy.

Explanation: Switching from unhealthy to longevity-based diet adds 5.4 to 10.8 years depending on gender.

- Add +10.8 years (males), +10.4 years (females) for healthy diet
- Add +5.4 years for partial adherence

Source: https://www.nature.com/articles/s43016-023-00868-w

Query:

```
UPDATE dataset
SET life_expectancy = life_expectancy + 10.8
WHERE diet_type = 'Longevity' AND gender = 'Male';
UPDATE dataset
SET life_expectancy = life_expectancy + 10.4
WHERE diet_type = 'Longevity' AND gender = 'Female';
UPDATE dataset
SET life_expectancy = life_expectancy + 5.4
WHERE diet_type = 'Moderate';
```

Step 8: Smoking

Rule: Life expectancy decreases with smoking intensity.

Explanation:

- Heavy smokers lose ~13 years
- Moderate smokers lose ~9 years

- Light smokers lose ~5 years

Source: https://www.cbs.nl/en-gb/news/2017/37/heavy-smokers-cut-their-lifespan-by-13-years-on-average

Query:

```
UPDATE dataset
SET life_expectancy = life_expectancy - 13
WHERE smoking_status = 'Heavy';
UPDATE dataset
SET life_expectancy = life_expectancy - 9
WHERE smoking_status = 'Moderate';
UPDATE dataset
SET life_expectancy = life_expectancy - 5
WHERE smoking_status = 'Light';
```

Step 9: Alcohol Consumption

Rule: Life expectancy reduces with increased alcohol intake.

Explanation:

- 7–14 drinks/week: -0.5 years
- 14–25 drinks/week: -1 to -2 years
- 25 drinks/week: -4 to -5 years

Source: https://www.health.harvard.edu/blog/sorting-out-the-health-effects-of-alcohol-2018080614427

Query:

```
UPDATE dataset
SET life_expectancy = life_expectancy - 0.5
WHERE alcohol_consumption BETWEEN 7 AND 14;
UPDATE dataset
SET life_expectancy = life_expectancy - 1.5
WHERE alcohol_consumption BETWEEN 15 AND 25;
UPDATE dataset
SET life_expectancy = life_expectancy - 4.5
WHERE alcohol_consumption > 25;
```

Step 10: BMI (Body Mass Index)

Rule: Underweight and obese individuals have reduced life expectancy.

Explanation:

- Obese men: -4.2 years
- Obese women: -3.5 years

- Underweight men: -4.3 years
- Underweight women: -4.5 years

Source: https://www.thelancet.com/journals/landia/article/PIIS2213-8587(18)30288-2/fulltext

Query:

1. UPDATE dataset
2. SET life_expectancy = life_expectancy - 4.2
3. WHERE bmi >= 30 AND gender = 'Male';
4. UPDATE dataset
5. SET life_expectancy = life_expectancy - 3.5
6. WHERE bmi >= 30 AND gender = 'Female';
7. UPDATE dataset
8. SET life_expectancy = life_expectancy - 4.3
9. WHERE bmi < 18.5 AND gender = 'Male';
10. UPDATE dataset
11. SET life_expectancy = life_expectancy - 4.5
12. WHERE bmi < 18.5 AND gender = 'Female';

Step 11: Sleep Duration

Rule: Adequate sleep improves longevity.

Explanation:
Men with adequate sleep live ~5 years longer.

Source: https://mcpress.mayoclinic.org/healthy-aging/how-quality-sleep-impacts-your-lifespan/

Query:

1. UPDATE dataset
2. SET life_expectancy = life_expectancy + 5
3. WHERE sleep_hours_per_day BETWEEN 7 AND 9 AND gender = 'Male';

Step 12: Work Stress

Rule: Chronic work stress correlates with increased health risk.

Explanation: Stress is associated with increased smoking, alcohol use, and reduced longevity.

Source:
https://www.researchgate.net/publication/230564274_The_effect_of_job_stress_on_smoking_and_alcohol_consumption

Query:

1. UPDATE dataset
2. SET life_expectancy = life_expectancy - 2

3. WHERE work_stress_level = 'High';

Step 13: Social Life

Rule: Social engagement impacts mental and physical health.

Explanation:

- High social interaction: +0.2 years
- Low social interaction: -0.2 years

Source: https://pmc.ncbi.nlm.nih.gov/articles/PMC11026051/

Query:

1. UPDATE dataset
2. SET life_expectancy = life_expectancy + 0.2
3. WHERE social_life = 'High';

4. UPDATE dataset
5. SET life_expectancy = life_expectancy - 0.2
6. WHERE social_life = 'Low';

Step 14: Medical History Impact

Rule: Chronic or serious illness reduces life expectancy.

Condition	Life Expectancy Impact	Source
Asthma	-1.4 years	Link
Diabetes	-10 years	Link
Cancer (young)	-30 years	Link
HIV/AIDS	-30 years	Link
Heart Disease	-12 years	Link
Kidney/Liver	-7 years	Link

Table 1.2: Serious health illness with life expectancy impact

Query:

1. UPDATE dataset
2. SET life_expectancy = life_expectancy - 1.4
3. WHERE medical_history = 'Asthma';
4. UPDATE dataset
5. SET life_expectancy = life_expectancy - 10
6. WHERE medical_history = 'Diabetes';
7. UPDATE dataset
8. SET life_expectancy = life_expectancy - 30
9. WHERE medical_history IN ('Cancer', 'HIV/AIDS');
10. UPDATE dataset
11. SET life_expectancy = life_expectancy - 12
12. WHERE medical_history = 'Heart Disease';

```
UPDATE dataset
SET life_expectancy = life_expectancy - 7
WHERE medical_history IN ('Kidney Disease', 'Liver Disease');
```

B. Reproducibility and Model Configuration

To ensure transparency and reproducibility, all components of the AI recommendation system were built using well-documented open-source tools and maintained in a version-controlled environment. The language model used for generating health recommendations was Mistral-7B-Instruct-v0.3, a quantized, instruction-tuned variant available via Hugging Face. This model was not fine-tuned further but used in its pre-trained form for efficient inference.

The embedding model used for semantic search was sentence-transformers/all-mpnet-base-v2, while vector storage and retrieval were handled using Pinecone. The Pinecone index was configured as a pod_type="p1.x1" with a cosine distance metric and 384-dimensional vectors. All embeddings were stored in a single namespace to support efficient contextual retrieval.

For the conversational retrieval chain, LangChain's ConversationalRetrievalChain was used with memory enabled to maintain chat context. The inference pipeline was executed using Hugging Face's InferenceClient, ensuring compatibility with quantized transformer models.

Reproducibility was ensured by freezing all package versions using a requirements.txt file and running experiments in a controlled Python environment (Python 3.10). A full export of the source code, model configurations, and input data formats is publicly available on the GitHub repository:

https://github.com/HardikSen16/Health-wise-Lifespan-Assessment-using-IoT-and-AI

Component	**Configuration / Version**
Language Model	mistralai/Mistral-7B-Instruct-v0.3 (quantized)
Embedding Model	sentence-transformers/all-mpnet-base-v2
Pinecone Index	cosine metric, 384-d, p1.x1 pod
Transformers Library	transformers==4.40.1
Sentence Transformers	sentence-transformers==2.6.1
Pinecone Client	pinecone-client==3.0.0
LangChain	langchain==0.1.14
Hugging Face Hub	huggingface-hub==0.20.3
Python Version	Python 3.10.13
GitHub Repository	Project Repository

Table 1.3: Software and Model Configuration

PROJECT DETAILS

Student Details			
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Project Details			
Project Title	**Healthwise Lifespan Assessment Using IoT and AI**		
Project Duration	6 Months	Date of reporting	3rd Jan 2025
Organization Details			
Organization Name	**Manipal Institute of Technology, Manipal**		
Full postal address with pin code	Manipal Institute of Technology, Eshwar Nagar, Udupi - Karkala Rd, Manipal, Karnataka 576104		
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